Berry and Uhlmann Geometry of Entangled Photon-Added Coherent States under the Global Two-Mode Cycle

$U(\phi) = e^{-i(\hat{n}_A+\hat{n}_B)\phi}$

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Abstract

We develop the geometric-phase structure of bipartite photon-added coherent states (PACS) taking as the primary cycle the *free evolution of both modes*, $U(\phi) = e^{-i(\hat{n}_A+\hat{n}_B)\phi}$. This is the evolution an actual propagating two-mode field undergoes, and it is a more natural starting point than a cycle acting on one mode alone: under the global cycle *both* reduced states traverse isospectral orbits and *both* carry Uhlmann holonomy, so the two subsystems are treated on an equal footing from the outset. We obtain in closed form the global Berry phase, which is the total mean photon number and therefore splits exactly into single-mode contributions $\Phi_B = \Phi_B^{(A)} + \Phi_B^{(B)}$; in the balanced symmetric family it is an affine function of the concurrence and exactly twice the single-mode result. For the mixed reduced states we derive the Uhlmann holonomy from the parallel-transport condition in its correct anticommutator form, $[S, \dot{S}] = i\{\rho, \Lambda\}$, which yields the compact statement that parallel transport renormalizes the off-diagonal generator by exactly the concurrence, $(G+\Lambda)_{\rm od} = C\, G_{\rm od}$, and hence $\Phi_U = \mathrm{Arg}\{(-1)^\omega[\cos\pi b - i\,\omega(1-2p)b^{-1}\sin\pi b]\}$ with $b = \omega\sqrt{1+C^2-4p(1-p)}$. We compute the concurrence of the entangled PACS in closed form, $C = 2\mu\nu N_1N_2/\mathcal{N}^2$, obtained from $C = 2|\det M|$ with the branch overlaps cancelling identically; the coherent amplitude thus acts as a continuous knob taking the entanglement from maximal (orthogonal Fock branches at $\alpha \to 0$) to zero (parallel branches at large α). Expressing the holonomy through the concurrence gives two clean statements: in the balanced symmetric family $p = C/2$ exactly, so $b = \Delta\sqrt{C^2+(1-C)^2}$ and Φ_U is a pure function of C; while on the locus $p = 1/2$ where the singularities live, $b = \Delta C$ and Z_U becomes real, reducing Φ_U to a $\mathbb{Z}_2$ index valued in $\{0, \pi\}$. The nodal condition is then that the effective winding $\omega_{\rm eff} = \Delta C$ cross a half-integer, which quantizes the concurrence to a finite, equally spaced ladder: each photon-addition separation Δ supports exactly Δ isolated Uhlmann singularities at $C_k = (2k+1)/2\Delta$, $k = 0, \ldots, \Delta-1$, symmetric about $C = 1/2$, shared by both subsystems and all bounded by a critical amplitude α_g defined by $|g_{mn}(\alpha_g)|^2 = 1/2$. We further explain why the two subsystems are *not* Uhlmann-symmetric despite sharing a concurrence: Schmidt isospectrality fixes the length of each reduced Bloch vector but not its orientation relative to that mode's own logical axis, so the local populations p_A, p_B differ whenever $\beta \neq \alpha$; in the crossed family the winding numbers differ in sign instead, producing an exact antisymmetry $\Phi_U^B = -\Phi_U^A$. In balanced form these statements sharpen into $\sqrt{p_Ap_B} = C/2$: the concurrence fixes only the geometric mean of the two local populations, while their ratio $\eta = \sqrt{p_A/p_B}$ is free — and the physical range of η pinches to $\{1\}$ as $C \to 1$, so maximal entanglement enforces symmetry. We then compute the *interferometric* (Sjöqvist) phase of both subsystems and establish its gauge invariance. In the laboratory frame, where the cycle is a phase shifter, it collapses to $\Phi_s^{\rm int} = \mathrm{Arg}\sum_k \lambda_k e^{2\pi i\nu_k}$ — the Pancharatnam sum of the Berry phases $2\pi\nu_k$ of the Schmidt modes, weighted by their populations — and is invariant, term by term, under both the eigenvector rephasing and the residual freedom $U \to Ue^{i\Xi}$, $[\Xi, \rho] = 0$, that leaves the path of density matrices unchanged. Its modulus is the fringe contrast and is a direct function of the concurrence, $\mathcal{V}_s = \sqrt{1 - C^2 \sin^2 2\pi\delta_s}$, which inverts into a witness: *the interference contrast of one mode alone determines the concurrence of the global state*, verified to 3×10^{-14}. Comparing the two mixed-state phases on the same loop isolates their difference to a single statement about parallel transport in the eigenbasis of ρ: Uhlmann keeps the diagonal part of the generator and rescales the off-diagonal part by C, while the interferometric prescription deletes the diagonal part and keeps the off-diagonal part intact. The interpolation $G_{\text{eff}} = G_d + sG_{\text{od}}$ gives $b_s = \omega s$ on $p = 1/2$, so all nodes lie on the lines $s = (2k+1)/2\Delta$: the Uhlmann phase runs along $s = C$ and meets every one of them, whereas the interferometric phase runs along $s = 0$ and meets none — indeed $|Z^{\text{int}}| \geq \sqrt{1 - C^2} > 0$, its only singularity being the maximally entangled point where the gauge group grows from $U(1) \times U(1)$ to $U(2)$. Every closed form is validated against independent numerics: direct summation in a truncated two-mode Fock space for the Berry sector, two connection-free holonomy algorithms for the Uhlmann sector, and the unevaluated definition of the interferometric phase for the third. A fully commented Jupyter notebook reproduces all nineteen figures and all four tables.

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1 Introduction

Photon addition turns a Glauber coherent state into a strongly nonclassical one while preserving an essentially closed analytic structure controlled by a single integer. The photon-added coherent states (PACS) of Agarwal and Tara [1] have since become a standard laboratory for the classical–quantum boundary of light. Entanglement in bipartite PACS was analysed by Domínguez-Serna, Mendieta-Jiménez and Rojas [5], who showed that superpositions of *nonorthogonal* PACS branches reduce, after Gram–Schmidt orthogonalization, to an effective two-qubit problem with a computable Wootters concurrence [4]. In parallel, geometric phases of coherent-state superpositions have been studied for two-mode entangled coherent states [6] and for nonlinear and squeezed states [7], while Uhlmann's mixed-state holonomy [3] has been developed for coherent states by Wang *et al.* [8].

Why the global cycle. A geometric phase is a property of a *path*, so the first thing to fix is which path. Earlier treatments of this problem cycled a single mode, $U_A(\phi) \otimes I_B$. That choice has a structural drawback: by the cyclicity of the partial trace, a strictly local unitary leaves ρ_B *exactly* invariant,

$$\mathrm{Tr}_A\left[(U_A \otimes I)\rho_{AB}(U_A^\dagger \otimes I)\right] = \mathrm{Tr}_A[\rho_{AB}], \tag{1}$$

which is the content of no-signalling. The path of ρ_B is then a *point*, its Uhlmann holonomy is the identity, and the second subsystem drops out of the discussion — not because it lacks geometry, but because the chosen cycle does not probe it.

Here we take instead the *global* cycle

$$U(\phi) = e^{-i(\hat{n}_A+\hat{n}_B)\phi} = U_A(\phi) \otimes U_B(\phi), \qquad \hat{n}_s = a_s^\dagger a_s, \tag{2}$$

i.e. free propagation of the two-mode field. It is the physically natural evolution, and under it $\rho_A(\phi) = U_A\rho_A(0)U_A^\dagger$ and $\rho_B(\phi) = U_B\rho_B(0)U_B^\dagger$ both move: both subsystems traverse isospectral loops and both carry nontrivial Uhlmann holonomy. The whole analysis is rebuilt on this basis.

Contributions.

(i) The closed-form global Berry phase, equal to the total mean photon number, with its exact decomposition into single-mode contributions and its affine dependence on the concurrence (Sec. 4, Figs. 1–5).

(ii) The Uhlmann connection derived from the parallel-transport condition in anticommutator form, giving the compact result $(G + \Lambda)_{\rm od} = C\,G_{\rm od}$ and the closed phase with $b = \omega\sqrt{1 + C^2 - 4p(1-p)}$ (Sec. 6).

(iii) A finite nodal set: exactly Δ isolated singularities at $C_k = (2k + 1)/2\Delta$, common to both subsystems, all bounded by α_g (Sec. 7, Figs. 6, 8).

(iv) An explanation of the $A \leftrightarrow B$ asymmetry: the concurrence is shared but the local populations are not, with the symmetry locus $\beta = \alpha$ and an exact antisymmetry in the crossed family (Sec. 10, Figs. 7, 11).

(v) The concurrence of the entangled PACS in closed form and its limits (Sec. 5, Fig. 12); the Uhlmann phase written through the concurrence — a pure function of C in the balanced symmetric family, a $\mathbb{Z}_2$ index on $p = 1/2$ (Sec. 8, Figs. 13–14) — and the resulting quantization of the concurrence at the singular points (Sec. 9).

(vi) A complete independent validation layer and a reproducible notebook (Sec. 13).

2 Photon-added coherent states

A PACS is obtained by applying the creation operator n times to a coherent state and renormalizing, $|\alpha, n\rangle = a^{\dagger n}|\alpha\rangle/[\langle\alpha|\,a^n a^{\dagger n}\,|\alpha\rangle]^{1/2}$. Inserting $|\alpha\rangle = e^{-|\alpha|^2/2}\sum_j \alpha^j/\sqrt{j!}\,|j\rangle$ and using $a^{\dagger n}|j\rangle = \sqrt{(j+n)!/j!}\;|j+n\rangle$ gives $a^{\dagger n}|\alpha\rangle = e^{-|\alpha|^2/2}\sum_{k\geq 0}\alpha^k\sqrt{(n+k)!}/k!\;|n+k\rangle$, whose norm is fixed by the PACS identity $\langle\alpha|\,a^n a^{\dagger n}\,|\alpha\rangle = n!\,L_n(-|\alpha|^2)$. Hence

$$|\alpha, n\rangle = \frac{e^{-|\alpha|^2/2}}{\sqrt{n!\,L_n(-|\alpha|^2)}}\sum_{k=0}^{\infty}\frac{\alpha^k\sqrt{(n+k)!}}{k!}\;|n+k\rangle\,, \tag{3}$$

with L_n the ordinary Laguerre polynomial. The mean photon number follows from the Laguerre recursion,

$$\bar{n}_{\alpha,n} = (n+1)\frac{L_{n+1}(-|\alpha|^2)}{L_n(-|\alpha|^2)} - 1, \tag{4}$$

which correctly reduces to $|\alpha|^2$ at $n = 0$.

Under the global cycle (2) each PACS transforms as

$$U(\phi)\,|\alpha, n\rangle = e^{-in\phi}\left|\alpha e^{-i\phi}, n\right\rangle, \tag{5}$$

because $e^{-i\hat{n}\phi}$ is diagonal in the Fock basis: the coherent amplitude rotates and the photon-addition index contributes an *explicit* phase $e^{-in\phi}$. These index phases are what generate the winding numbers below.

3 Bipartite model and logical-qubit reduction

Since PACS branches are not orthogonal, a bipartite superposition must be orthogonalized before it can be read as a two-qubit state. With p_1, p_2 the local branch overlaps and $N_j = \sqrt{1-|p_j|^2}$, the Gram–Schmidt bases are

$$|0_s\rangle = (\text{first branch}), \qquad |1_s\rangle = \frac{(\text{second branch}) - p_s\,|0_s\rangle}{N_s}, \qquad s = A, B, \tag{6}$$

and the reduced states become rank-2 with a Wootters concurrence $C = 2\sqrt{\det \rho_s}$ [4]. We use two families:

Branch-correlated.

$$|\Phi\rangle \propto \mu\,|\alpha, m\rangle_A\,|\beta, m\rangle_B + \nu e^{i\chi}\,|\alpha, n\rangle_A\,|\beta, n\rangle_B\,, \tag{7}$$

whose branch phases under (5) are $e^{-i2m\phi}$ and $e^{-i2n\phi}$; each mode separately winds with $\omega_A = \omega_B = +\Delta$, $\Delta \equiv n - m$.

Crossed (swap).

$$|\Phi\rangle \propto \mu\,|\alpha, m\rangle_A\,|\beta, n\rangle_B + \nu e^{i\chi}\,|\beta, n\rangle_A\,|\alpha, m\rangle_B\,. \tag{8}$$

Here the photon-addition indices are *interchanged* between modes, so $\omega_A = +\Delta$ while $\omega_B = -\Delta$: the two subsystems wind in opposite senses.

With $\nu = \sqrt{1-\mu^2}$, $g_\alpha \equiv g_{mn}(\alpha) = \langle \alpha, m|\alpha, n\rangle$ and $h_\beta \equiv g_{mn}(\beta)$, the branch-correlated family gives

$$\mathcal{N}^2 = 1 + 2\mu\nu g_\alpha h_\beta, \qquad p_A = \frac{\nu^2(1-g_\alpha^2)}{\mathcal{N}^2}, \qquad p_B = \frac{\nu^2(1-h_\beta^2)}{\mathcal{N}^2}, \tag{9}$$

$$\det \rho_A = \det \rho_B = \frac{\mu^2\nu^2(1-g_\alpha^2)(1-h_\beta^2)}{\mathcal{N}^4} = \left(\frac{C}{2}\right)^2, \tag{10}$$

while the crossed family, with $s \equiv \langle \alpha, m|\beta, n\rangle$, gives $\mathcal{N}^2 = 1 + 2\mu\nu s^2$, $p_A = p_B = \nu^2(1-s^2)/\mathcal{N}^2$ and $C = 2\mu\nu(1-s^2)/\mathcal{N}^2$; there $p_A = p_B$ holds identically because the two branches are exchanged copies of one another, so $N_1 = N_2$.

Equation (10) is worth emphasizing: for a bipartite *pure* state the Schmidt decomposition makes ρ_A and ρ_B isospectral, so the determinant — and hence the concurrence — is *shared* by the two subsystems. This will be central in Sec. 10.

4 Global Berry phase

The Berry connection of the global cycle is the total mean photon number,

$$\mathcal{A}_\phi = \langle \Phi|\,(\hat{n}_A + \hat{n}_B)\,|\Phi\rangle\,, \tag{11}$$

independent of ϕ since the generator commutes with itself, so $\Phi_B = \int_0^{2\pi} \mathcal{A}_\phi\, d\phi = 2\pi\langle \hat{n}_A + \hat{n}_B\rangle$. Expanding (7) and using $\kappa_\alpha \equiv \langle \alpha, m|\, a^\dagger a\,|\alpha, n\rangle$ (Appendix B),

$$\boxed{\frac{\Phi_B}{2\pi} = \frac{\mu^2(\bar{n}_{\alpha,m} + \bar{n}_{\beta,m}) + \nu^2(\bar{n}_{\alpha,n} + \bar{n}_{\beta,n}) + 2\mu\nu\cos\chi\,(\kappa_\alpha h_\beta + g_\alpha \kappa_\beta)}{1 + 2\mu\nu\cos\chi\, g_\alpha h_\beta}} \tag{12}$$

Exact mode decomposition. Because the generator is a *sum* of commuting single-mode operators, the phase splits exactly,

$$\Phi_B = \Phi_B^{(A)} + \Phi_B^{(B)}, \qquad \frac{\Phi_B^{(A)}}{2\pi} = \langle \hat{n}_A \rangle, \quad \frac{\Phi_B^{(B)}}{2\pi} = \langle \hat{n}_B \rangle, \tag{13}$$

with $\langle \hat{n}_A \rangle = [\mu^2 \bar{n}_{\alpha,m} + \nu^2 \bar{n}_{\alpha,n} + 2\mu\nu \cos\chi\, \kappa_\alpha h_\beta]/\mathcal{N}^2$ and the mirror expression for B. This decomposition, verified to 10^{-16} in Sec. 13, is displayed in Fig. 5: for $\beta \neq \alpha$ the two modes contribute unequally, and the mode whose amplitude is held fixed contributes a nearly constant amount.

Symmetric family and concurrence dependence. For $\beta = \alpha$ the two contributions coincide and Φ_B is exactly *twice* the single-mode result. In the balanced symmetric case ($\mu = \nu$, $\chi = 0$), writing $g^2 = (1 - C)/(1 + C)$,

$$\frac{\Phi_B}{2\pi} = (1 + C)\, \bar{n}_{\mathrm{av}} + (1 - C)\, Q_{mn}, \qquad \bar{n}_{\mathrm{av}} = \tfrac{1}{2}(\bar{n}_{\alpha,m} + \bar{n}_{\alpha,n}), \tag{14}$$

again *affine in the concurrence*, interpolating between the mean-photon average at $C = 0$ and the interference term Q_{mn} at $C = 1$, and again exactly twice the one-mode expression.

Branch-phase interference. The cycle in ϕ produces a connection that already contains the coherent cross term κg, yet no oscillation in ϕ: the interference is real but hidden inside a constant. Introducing the tunable relative branch phase χ of Eqs. (7) exposes it as an explicit 2π-periodic modulation, Fig. 3.

5 Concurrence of the entangled PACS

Because the concurrence controls both the Berry phase (through Eq. (14)) and the entire Uhlmann sector below, we derive it explicitly.

Step 1: logical basis. Inverting Eq. (6),

$$|\alpha, n\rangle = g_\alpha |0_A\rangle + N_1 |1_A\rangle, \qquad |\beta, n\rangle = h_\beta |0_B\rangle + N_2 |1_B\rangle, \tag{15}$$

with $N_1 = \sqrt{1 - g_\alpha^2}$, $N_2 = \sqrt{1 - h_\beta^2}$.

Step 2: coefficient matrix. Substituting into Eq. (7) at $\chi = 0$,

$$|\Phi\rangle = \frac{1}{\mathcal{N}} \Big[(\mu + \nu g_\alpha h_\beta) |0_A 0_B\rangle + \nu g_\alpha N_2 |0_A 1_B\rangle + \nu N_1 h_\beta |1_A 0_B\rangle + \nu N_1 N_2 |1_A 1_B\rangle \Big], \tag{16}$$

i.e. $|\Phi\rangle = \sum_{ij} M_{ij} |i_A j_B\rangle$ with

$$M = \frac{1}{\mathcal{N}} \begin{pmatrix} \mu + \nu g_\alpha h_\beta & \nu g_\alpha N_2 \\ \nu N_1 h_\beta & \nu N_1 N_2 \end{pmatrix}. \tag{17}$$

Step 3: concurrence. For a pure two-qubit state $C = 2|\det M|$, and the overlap terms cancel identically:

$$\det M = \frac{\nu N_1 N_2}{\mathcal{N}^2}\left[\mu + \nu g_\alpha h_\beta - \nu g_\alpha h_\beta\right] = \frac{\mu\nu N_1 N_2}{\mathcal{N}^2}, \tag{18}$$

so that

$$\boxed{C = \frac{2\mu\nu N_1 N_2}{\mathcal{N}^2} = \frac{2\mu\nu\sqrt{(1-g_\alpha^2)(1-h_\beta^2)}}{1+2\mu\nu\, g_\alpha h_\beta}} \tag{19}$$

consistent with $\det\rho_A = \det\rho_B = |\det M|^2 = (C/2)^2$, Eq. (10). Substituting the closed-form overlap (76) renders C an explicit function of $(\mu,\alpha,\beta;m,n)$ alone.

Limits. The overlap controls everything (Fig. 12):

- $\alpha,\beta\to 0$: $g\to 0$, since the PACS degenerate into the Fock states $|m\rangle, |n\rangle$, which *are* orthogonal. Then $C\to 2\mu\nu$, maximal for the given weights; balanced gives $C=1$.
- $\alpha,\beta\to\infty$: $g\to 1$, the two branches become parallel, and $C\to 0$: a product state.

The coherent amplitude is thus a continuous knob taking the entanglement from maximal to zero, with nonorthogonality the precise mechanism that degrades it. In the balanced symmetric family Eq. (19) collapses to

$$C = \frac{1-q}{1+q}, \qquad q = |g_{mn}(\alpha)|^2. \tag{20}$$

6 Uhlmann holonomy of the reduced states

Under the global cycle both reduced states follow isospectral orbits

$$\rho_s(\phi) = V_s(\phi)\rho_s(0)V_s^\dagger(\phi), \qquad V_s(\phi) = e^{i\frac{\omega_s}{2}\sigma_z\phi}, \qquad s = A,B. \tag{21}$$

This deserves a derivation, because the physical generator is $\hat{n}_s$ — an unbounded operator with no σ_z in sight — and the reduction to Eq. (21) is where both the winding number ω_s and a frame choice enter.

6.1 Where σ_z comes from

Step 1: the cycle acting on one branch. Since $[\hat{n}, a^\dagger] = a^\dagger$ we have $e^{-i\hat{n}\phi}a^\dagger e^{i\hat{n}\phi} = a^\dagger e^{-i\phi}$ and $e^{-i\hat{n}\phi}|\alpha\rangle = |\alpha e^{-i\phi}\rangle$; the normalization is untouched because $L_k(-|\alpha e^{-i\phi}|^2) = L_k(-|\alpha|^2)$. Hence

$$e^{-i\hat{n}\phi}|\alpha,k\rangle = e^{-ik\phi}\left|\alpha e^{-i\phi},k\right\rangle. \tag{22}$$

Two distinct things happen: a c-number phase $e^{-ik\phi}$ that *carries the photon-addition index*, and a rigid rotation of the coherent amplitude $\alpha\to\alpha e^{-i\phi}$ that is the *same* for both branches and closes at $\phi = 2\pi$.

Step 2: the co-moving frame absorbs the amplitude rotation. ρ_s has rank two. Take the Gram–Schmidt basis (6) at $\phi = 0$ and transport it, attaching to each logical direction the phase of its own branch,

$$|e_j(\phi)\rangle = e^{+ik_j\phi}\, e^{-i\hat{n}\phi}|j\rangle, \qquad (k_0,k_1) = (m,n). \tag{23}$$

This is not an arbitrary choice: $|e_0(\phi)\rangle = |\alpha_\phi, m\rangle$ with $\alpha_\phi = \alpha e^{-i\phi}$, and $|e_1(\phi)\rangle$ is exactly the *instantaneous* Gram–Schmidt vector of the rotated branches, because the overlap rotates too,

$\langle\alpha_\phi, m|\alpha_\phi, n\rangle = e^{i\Delta\phi} g_{mn}(\alpha)$, while its modulus stays fixed. The frame is orthonormal for every ϕ and closes, $|e_j(2\pi)\rangle = |j\rangle$. In it,

$$\tilde{\rho}_{jk}(\phi) = e^{-ik_j\phi}\rho_{jk}(0)e^{+ik_k\phi} = \big[D^\dagger(\phi)\,\rho(0)\,D(\phi)\big]_{jk}, \qquad D(\phi) = \mathrm{diag}\big(e^{ik_0\phi}, e^{ik_1\phi}\big). \tag{24}$$

The amplitude rotation has gone with the frame — it was pure transport — and all that survives is the diagonal matrix of branch phases.

Step 3: the trace part is gauge. Any diagonal 2×2 unitary splits into its trace and traceless pieces,

$$D(\phi) = e^{i\bar{k}\phi}\, e^{-i\frac{\Delta}{2}\sigma_z\phi}, \qquad \bar{k} = \frac{k_0 + k_1}{2}, \qquad \Delta = k_1 - k_0, \tag{25}$$

and the scalar $e^{i\bar{k}\phi}\mathbb{1}$ cancels identically in $D^\dagger\rho D$. What remains is Eq. (21) with $\omega_s = k_1^{(s)} - k_0^{(s)}$.

Why σ_z, and why Δ. σ_z because the cycle is the free evolution and therefore *does not mix the branches* — each is an eigensector of the addition index — so its generator is diagonal in the branch labels, and a diagonal Hermitian 2×2 matrix has only the two pieces $a_0\mathbb{1} + a_z\sigma_z$. And Δ rather than n because ρ is blind to $\mathbb{1}$: the mean $\bar{k}$ does not vanish from the problem, it reappears in the *Berry* phase of the global pure state, $\Phi_B/2\pi = \langle\hat{n}_A + \hat{n}_B\rangle$, which is sensitive to the overall phase. Mixed-state geometry of a reduced state sees only the difference. Reading off the branch pairs of each mode gives the winding numbers quoted in Sec. 3:

family	branches of A	branches of B	ω_A	ω_B
branch-correlated	(m, n)	(m, n)	$+\Delta$	$+\Delta$
crossed	(m, n)	(n, m)	$+\Delta$	$-\Delta$

The exact antisymmetry (49) of the crossed family originates precisely in that reversal of order: $k_1 - k_0 = m - n = -\Delta$.

The frame is part of the statement. The winding ω_s is not a property of the path $\phi \mapsto \rho_s(\phi)$ alone; it is a property of that path *together with* the frame (23). Transport the basis with the bare physical unitary, $|j(\phi)\rangle = e^{-i\hat{n}\phi}\,|j\rangle$, and the matrix of ρ_s is *constant*: no winding and no holonomy at all. The co-moving frame is the one that pins the coherent amplitude and lets the relative branch phase run, and it is the natural choice for the reduced geometry. It is also exactly the distinction that reappears in Sec. 12: the laboratory interferometric phase uses the bare phase shifter $e^{-i\hat{n}_s\phi}$, the logical one uses Eq. (23), and the two differ by the offset $\pi\,\mathrm{Tr}\,\mathcal{N}_s$.

Equations (23)–(24) are checked directly against the exact truncated-Fock reduced states: the 2×2 block reproduces $[\,[1-p,\ c_0e^{i\omega\phi}], [c_0e^{-i\omega\phi},\ p]\,]$ to 9×10^{-16}, with the winding recovered as $+2, +2$ (correlated) and $+3, -3$ (crossed), Table 2.

6.2 The parallel-transport condition

With $\rho = WW^\dagger$, $W = \sqrt{\rho}\,U$, $S \equiv \sqrt{\rho}$ and $\dot{U} = -i\Lambda U$, one has $W^\dagger\dot{W} = U^\dagger(S\dot{S} - i\rho\Lambda)U$. Uhlmann's condition is that $W^\dagger\dot{W}$ be Hermitian, which gives

$$\boxed{[S, \dot{S}] = i\,\{\rho, \Lambda\}} \tag{26}$$

an *anticommutator* (Sylvester) equation. In the eigenbasis of ρ, with eigenvalues λ_j, it is solved elementwise by

$$\Lambda_{jk} = \frac{-i\,[S,\dot{S}]_{jk}}{\lambda_j + \lambda_k}. \tag{27}$$

We stress that the superficially similar sandwich expression $\Lambda = \frac{1}{2i}S^{-1}[S,\dot{S}]S^{-1}$, whose elements carry a denominator $2\sqrt{\lambda_j\lambda_k}$, does *not* solve Eq. (26) unless $\lambda_j = \lambda_k$; the two coincide only for the maximally mixed state. Section 13 confirms Eq. (27) against two connection-free numerical holonomies.

6.3 Closed form

For the isospectral loop, $\dot{S} = i[G,S]$ with $G = \frac{\omega}{2}\sigma_z$, so $[S,\dot{S}]_{jk} = -iG_{jk}(\sqrt{\lambda_j} - \sqrt{\lambda_k})^2$ and Eq. (27) gives a purely off-diagonal connection

$$\Lambda_{jk} = -G_{jk}\frac{(\sqrt{\lambda_j} - \sqrt{\lambda_k})^2}{\lambda_j + \lambda_k}, \qquad j \neq k. \tag{28}$$

Since $\operatorname{Tr}\rho = 1$ we have $\lambda_j + \lambda_k = 1$ for $j \neq k$, and $2\sqrt{\lambda_+\lambda_-} = 2\sqrt{\det\rho} = C$, so $(\sqrt{\lambda_+} - \sqrt{\lambda_-})^2 = 1 - C$ and the connection collapses to

$$\boxed{\Lambda_{\mathrm{od}} = -(1-C)\,G_{\mathrm{od}} \qquad \Longrightarrow \qquad (G+\Lambda)_{\mathrm{od}} = C\,G_{\mathrm{od}}} \tag{29}$$

the concurrence is exactly the factor by which parallel transport renormalizes the off-diagonal part of the generator. Writing the Bloch vector of $\rho_s(0)$ with $\cos\theta_s = (1-2p_s)/r$, $r = \sqrt{1-C^2}$, and using $e^{iG2\pi} = (-1)^{\omega}\mathbb{1}$ for integer ω, the holonomy $U_{\mathrm{hol}} = e^{iG2\pi}e^{-i(G+\Lambda)2\pi}$ gives

$$\boxed{\Phi_U = \operatorname{Arg}\left\{(-1)^{\omega}\left[\cos(\pi b) - i\,\frac{\omega(1-2p)}{b}\sin(\pi b)\right]\right\}, \qquad b = \omega\sqrt{1 + C^2 - 4p(1-p)}} \tag{30}$$

where the compact b follows from $b^2 = \omega^2(\cos^2\theta + C^2\sin^2\theta)$ on substituting $r^2 = 1 - C^2$. The prefactor $(-1)^{\omega}$ comes from $e^{iG2\pi}$, which is a *scalar*, and therefore multiplies the whole bracket; attaching it to the cosine alone conjugates Z_U for odd ω. The modulus $|Z_U|$ is insensitive to that placement — so the nodal set, the $\mathbb{Z}_2$ staircase and the quantization of C below are unaffected — but the sign of Φ_U for odd ω is not. Section 13 therefore compares the full complex Z_U, not merely its modulus, against the two connection-free holonomies. Two limits check the result: if G commutes with ρ the state does not move and $\Phi_U = 0$; and the maximally mixed loop is again trivial. At the population-symmetric point,

$$p = \tfrac{1}{2} \qquad \Longrightarrow \qquad b = \omega\,C, \tag{31}$$

so b is there *proportional* to the concurrence.

7 Nodal singularities

Define the complex Uhlmann amplitude $Z_U = (-1)^{\omega}[\cos(\pi b) - i\,\omega(1-2p)b^{-1}\sin(\pi b)]$, so $\Phi_U = \operatorname{Arg} Z_U$. A genuine phase singularity needs both parts to vanish; since $\cos(\pi b) = 0 \Rightarrow \sin(\pi b) \neq 0$, the factor $(1-2p)$ must vanish independently. Hence

$$\boxed{p = \tfrac{1}{2}, \qquad |b| = k + \tfrac{1}{2}, \quad k \in \mathbb{Z}_{\geq 0}} \tag{32}$$

At $p = 1/2$, Eq. (31) gives $|b| = \Delta\, C$, so the concurrence is quantized to

$$\boxed{C_k = \frac{2k+1}{2\Delta}, \qquad k = 0, 1, \ldots, \Delta - 1} \tag{33}$$

the upper bound enforcing $C_k \leq 1$. Thus **each photon-addition separation Δ supports exactly Δ Uhlmann singularities**, equally spaced in concurrence by $1/\Delta$, isolated, with no accumulation point. Because only $|\omega_s|$ enters and the concurrence is shared, the set (33) is *the same for both subsystems*: which entanglement values are geometrically singular is a property of the pair.

7.1 Nodal coordinates and the critical amplitude

For the symmetric family, $p = 1/2$ in Eq. (9) solves exactly: $\mu = \nu(1 - 2q)$ with $q = g_{mn}(\alpha)^2$, i.e.

$$\mu(q) = \frac{1 - 2q}{\sqrt{1 + (1 - 2q)^2}}, \qquad \text{along which} \qquad C = 1 - 2q. \tag{34}$$

These relations are purely kinematic — properties of the state, not of the connection. Combining with Eq. (33),

$$q_k = \frac{1 - C_k}{2}, \qquad \mu_k = \frac{C_k}{\sqrt{1 + C_k^2}}, \qquad |g_{mn}(\alpha_k)|^2 = q_k, \tag{35}$$

with α_k obtained by root-finding on the monotonic function $|g_{mn}(\alpha)|^2$. Defining the critical amplitude by

$$|g_{mn}(\alpha_g)|^2 = \tfrac{1}{2}, \tag{36}$$

the $p = 1/2$ curve exists only for $\alpha \leq \alpha_g$. Since $C_k \geq 1/(2\Delta) > 0$ we have $q_k < 1/2$ strictly, so *every* node satisfies $\alpha_k < \alpha_g$: the critical amplitude is a genuine **bound** on the nodal region, but — the set being finite — not an accumulation point. Table 4 lists the complete nodal set for representative pairs, each verified to satisfy $p = 1/2$, $|b| = k + 1/2$ and $|Z_U| \lesssim 10^{-14}$ simultaneously.

7.2 The same construction for subsystem B

Everything above applies verbatim to B with $p_A \to p_B$ and $\omega_A \to \omega_B$; since only $|\omega_s| = \Delta$ enters and the concurrence is shared, the ladder (33) is identical on both sides. What differs is the locus on which the nodes sit. For $\beta \neq \alpha$ the condition $p_B = 1/2$ reads, from Eq. (9) with $G \equiv g_{mn}(\alpha)$, $H \equiv h_{mn}(\beta)$,

$$2(1 - \mu^2)(1 - H^2) = 1 + 2\mu\sqrt{1 - \mu^2}\, G\, H, \tag{37}$$

to be solved for μ at each α; setting $H = G$ recovers Eq. (34), as it must. Evaluating the difference of the two sides at the endpoints gives $1 - 2H^2$ at $\mu = 0$ and -1 at $\mu = 1$, so a root in $(0, 1)$ exists if and only if

$$|h_{mn}(\beta)|^2 < \tfrac{1}{2} \qquad \Longleftrightarrow \qquad \beta < \alpha_g(m, n), \tag{38}$$

independently of α. This is a structural contrast with A, whose criterion $\alpha \leq \alpha_g$ bounds its nodes from above: the B criterion constrains the *other* amplitude, leaving its own nodes free to sit well beyond α_g, as Figs. 8 and 16 show.

8 The Uhlmann phase as a function of the concurrence

Equation (30) mixes the concurrence with the population. We now identify the loci on which Φ_U becomes a function of C *alone*.

8.1 (C, η) coordinates: what entanglement fixes and what it does not

In the balanced correlated family ($\mu = \nu = 1/\sqrt{2}$) an exact identity ties both subsystems to the shared concurrence:

$$\boxed{\sqrt{p_A p_B} = \frac{C}{2}} \tag{39}$$

Introducing the *local asymmetry*

$$\eta \equiv \sqrt{\frac{p_A}{p_B}} = \sqrt{\frac{1-g_\alpha^2}{1-h_\beta^2}}, \qquad p_A = \frac{C}{2}\eta, \qquad p_B = \frac{C}{2\eta}, \tag{40}$$

separates the two roles cleanly: *the concurrence fixes the geometric mean of the two local populations, while η is the local degree of freedom that entanglement leaves free.* The $A \leftrightarrow B$ symmetry of Sec. 10 is exactly $\eta = 1$, i.e. $\beta = \alpha$.

Physical region. Requiring $p(1-p) \geq \det\rho = C^2/4$ on both sides gives $C\eta^2 - 2\eta + C \leq 0$, i.e.

$$\eta \in \left[\frac{1-r}{C}, \frac{1+r}{C}\right], \qquad r = \sqrt{1-C^2}, \tag{41}$$

an interval symmetric under $\eta \to 1/\eta$ (the product of its endpoints is exactly 1). It narrows as C grows, and

$$\boxed{C \to 1 \implies \eta \to 1} \tag{42}$$

maximal entanglement forces the two subsystems to be symmetric: at $C = 1$ one necessarily has $p_A = p_B = 1/2$, hence $\Phi_U^A = \Phi_U^B$ (and in fact $b = \Delta$, an integer, so $\Phi_U = 0$). Asymmetry is a luxury only partial entanglement can afford. See Fig. 14.

8.2 Balanced symmetric family: a pure function of C

With $\beta = \alpha$ and $\mu = \nu$ one has $\eta = 1$ and therefore $p = C/2$ exactly, so $1 + C^2 - 4p(1-p) = C^2 + (1-C)^2$ and

$$\boxed{b = \Delta\sqrt{C^2 + (1-C)^2}, \qquad \Phi_U = \mathrm{Arg}\left\{(-1)^\Delta \left[\cos(\pi b) - i\frac{\Delta(1-C)}{b}\sin(\pi b)\right]\right\}} \tag{43}$$

a function of C and Δ *only*, plotted in Fig. 13(a). Its range is $b \in [\Delta/\sqrt{2}, \Delta]$, minimal at $C = 1/2$ and equal to Δ at both endpoints $C \to 0$ and $C \to 1$.

8.3 On $p = 1/2$: a $\mathbb{Z}_2$ index

On the locus where the nodes live the imaginary part of Z_U vanishes identically (it carries the factor $1 - 2p$) and $b = \Delta C$, so

$$Z_U = (-1)^\Delta \cos(\pi\Delta C) \in \mathbb{R} \qquad \implies \qquad \boxed{\Phi_U \in \{0, \pi\}} \tag{44}$$

The Uhlmann phase ceases to be continuous and becomes a $\mathbb{Z}_2$ index determined by the concurrence alone: 0 where $(-1)^\Delta \cos(\pi\Delta C) > 0$ and π where it is negative. The nodes are precisely the domain walls between the two values, Fig. 13(c).

9 Quantization of the concurrence at the singular points

On $p = 1/2$ the parameter $b = \Delta C$ acts as an *effective winding number* that the concurrence tunes continuously,

$$\omega_{\text{eff}} \equiv b\big|_{p=1/2} = \Delta\, C \quad \text{sweeping } [0,\Delta] \text{ as } C \in [0,1], \tag{45}$$

and the nodal condition $|b| = k+\frac{1}{2}$ is the statement that this effective winding crosses a *half-integer*:

$$\boxed{\Delta\, C_k = k + \tfrac{1}{2} \qquad \Longleftrightarrow \qquad C_k = \frac{2k+1}{2\Delta}, \qquad k = 0, 1, \ldots, \Delta - 1} \tag{46}$$

Five consequences follow immediately.

(1) **Count.** Since $C \le 1$ only $k \le \Delta - 1$ fits: there are *exactly* Δ singularities. The photon-addition separation $\Delta = n - m$ literally counts the singular points.

(2) **Uniform spacing.** $C_{k+1} - C_k = 1/\Delta$: the singular concurrences are equidistant, being the *odd* multiples of $1/2\Delta$.

(3) **Reflection symmetry.** $C_k + C_{\Delta-1-k} = 1$, so the ladder is symmetric about $C = 1/2$.

(4) $\mathbb{Z}_2$ **jumps.** By Eq. (44), crossing each C_k flips Φ_U between 0 and π; there are exactly Δ flips between $C = 0$ and $C = 1$.

(5) **Shared by both subsystems.** Only $|\omega_s| = \Delta$ enters and the concurrence is shared, so A and B have the *same* ladder $\{C_k\}$ — though not the same coordinates (μ, α), as Fig. 8 shows.

Translated back to parameter space, on $p = 1/2$ the symmetric family gives $C = 1 - 2q$, so each C_k fixes an overlap $q_k = (1 - C_k)/2$ and with it an amplitude α_k; since $C_k \ge 1/(2\Delta) > 0$ one always has $q_k < 1/2$ and hence $\alpha_k < \alpha_g$, recovering the bound of Sec. 7.

10 Why the two subsystems are not Uhlmann-symmetric

Since the global state is pure, ρ_A and ρ_B are isospectral and share a concurrence. It is then natural to ask why $\Phi_U^A \neq \Phi_U^B$. The answer is that *the Uhlmann phase is not a function of the entanglement alone*. It depends on three inputs,

$$\Phi_U^s = \Phi_U(C,\ p_s,\ \omega_s), \tag{47}$$

of which only the first is symmetric:

ingredient	symmetric under $A \leftrightarrow B$?	reason
C, $\det\rho$, r	**yes**	Schmidt: identical spectra
p_s	**no**	not an eigenvalue, but a matrix element in the *local* basis
ω_s	**no** (crossed family)	indices interchanged between modes

The decisive one is p_s. Isospectrality fixes the *eigenvalues* of ρ_s, but $p_s = (\rho_s)_{22}$ is not an eigenvalue: it is a matrix element in the Gram–Schmidt basis (6) built from *that mode's own* branches, whose overlap is g_α for A and h_β for B. Nothing forces $g_\alpha = h_\beta$.

Geometric reading. With $\cos\theta_s = (1 - 2p_s)/r$ one has the exact identity

$$b_s = \omega_s \sqrt{\cos^2\theta_s + C^2 \sin^2\theta_s}, \tag{48}$$

where θ_s is the *tilt* between the reduced Bloch vector of mode s and the σ_z axis of its own logical basis. Therefore:

Schmidt's theorem fixes the **length** of each reduced Bloch vector (r, common to both sides) but **not its orientation** relative to each mode's local axis. Uhlmann holonomy depends on that orientation.

Entanglement is symmetric; *local nonorthogonality is not.* Concretely, for $(m,n)=(0,2)$, $\alpha=1.1$, $\beta=0.8$, $\mu=0.6$ the two sides share $C=0.748027$ exactly while $p_A=0.4725$, $p_B=0.5263$, i.e. tilts $\theta_A=85.2^\circ$ and $\theta_B=94.6^\circ$. Symmetry is restored exactly when the two local geometries coincide, $g_\alpha=h_\beta$, i.e. $\beta=\alpha$ for real amplitudes; Fig. 11(c) shows that the symmetry locus in the (α,β) plane is precisely the diagonal.

Antisymmetry in the crossed family. There $p_A=p_B$ identically but $\omega_B=-\omega_A$, hence $b_B=-b_A$. Inspecting Eq. (30): the scalar $(-1)^{\omega_B}=(-1)^{\omega_A}$ and $\cos(\pi b_B)=\cos(\pi b_A)$, so the real parts agree; while $\omega_B/b_B=\omega_A/b_A$ but $\sin(\pi b_B)=-\sin(\pi b_A)$, so the imaginary parts are opposite. Therefore $Z_U^B=\overline{Z_U^A}$ and

$$\boxed{\Phi_U^B=-\,\Phi_U^A,\qquad \Phi_U^A+\Phi_U^B=0} \tag{49}$$

exactly, for every α,β,μ,m,n — verified to floating-point zero in Sec. 13 and shown in Fig. 7(c). The entangled pair then accumulates no *net* Uhlmann holonomy even though each half accumulates a large one. Since complex conjugation does not move zeros, the two subsystems also share their nodal set exactly in that family.

11 A case with many singularities: $\Delta=8$

Equation (46) says the number of singularities is *exactly* Δ, so producing many of them is simply a matter of raising the photon-addition separation. Take

$$(m,n)=(0,8)\qquad\Longrightarrow\qquad\Delta=8, \tag{50}$$

which gives **eight Uhlmann nodes in A and eight in B**, at

$$C_k=\tfrac{1}{16},\tfrac{3}{16},\tfrac{5}{16},\tfrac{7}{16},\tfrac{9}{16},\tfrac{11}{16},\tfrac{13}{16},\tfrac{15}{16}, \tag{51}$$

equally spaced by $1/8$ and symmetric about $C=1/2$, as Eq. (46) requires. Figure 15 shows the $\mathbb{Z}_2$ staircase now taking eight steps, the effective winding $b=\Delta C$ crossing eight half-integers, and the eight nodes strung along the $p=1/2$ curve in the (μ,α) plane.

11.1 A numerical remark: Kummer's transformation

This case demands large amplitudes — $\alpha_g(0,8)=9.2301$, and the B nodes reach $\alpha\sim10^2$ — where the direct form of the overlap overflows: ${}_1F_1\big(n+1;\Delta+1;\alpha^2\big)$ grows like e^{α^2} and exceeds double precision already at $\alpha\gtrsim27$. Kummer's transformation

$${}_1F_1(n+1;\Delta+1;z)=e^z\,{}_1F_1(-m;\Delta+1;-z) \tag{52}$$

removes the problem at its root: since $b-a=-m\le0$, the transformed hypergeometric is a *polynomial of degree* m, and the exponential cancels against the prefactor of Eq. (76), leaving $e^{-(\alpha-\beta)^2/2}\le1$. The overlap then reads

$$\langle\alpha,m|\beta,n\rangle=\frac{e^{-(\alpha-\beta)^2/2}\,\alpha^\Delta\,n!}{\Delta!\,\sqrt{m!\,n!\,L_m(-\alpha^2)L_n(-\beta^2)}}\,{}_1F_1(-m;\Delta+1;-\alpha\beta)\,, \tag{53}$$

numerically stable for *any* amplitude, and the same substitution applies to the ratio in Q_{mn}, Eq. (78). For $m = 0$ the overlap even becomes elementary,

$$g_{0,n}(\alpha) = \frac{\alpha^n}{\sqrt{n!\, L_n(-\alpha^2)}}. \tag{54}$$

11.2 Where the eight nodes sit

In the symmetric family $\beta = \alpha$ one has $p_A = p_B$, so the two subsystems share the *same* eight nodes, listed in the left half of Table 1; they occupy $\alpha_k \in [3.54, 8.80]$, all below $\alpha_g = 9.2301$ as Sec. 7 requires. Fixing $\beta \neq \alpha$ separates the two sets dramatically (Fig. 16 and the right half of Table 1): subsystem A keeps its eight nodes below α_g, whereas those of B — whose existence criterion (38) depends on β alone — spread out to $\alpha \approx 97$, more than ten times α_g. The same entangled state, the same ladder $\{C_k\}$, and two completely different nodal geometries.

Table 1: The eight Uhlmann nodes of the $\Delta = 8$ case, $(m, n) = (0, 8)$, $\alpha_g = 9.2301$. Left: symmetric family $\beta = \alpha$, where A and B share the nodes. Right: $\beta = 4$ fixed, where the two sets separate. Every entry satisfies $p = 1/2$ and $b = k + \frac{1}{2}$ with $|Z_U| < 5 \times 10^{-14}$.

		$\beta = \alpha$ (shared)		$\beta = 4$: subsystem A		$\beta = 4$: subsystem B	
k	C_k	μ_k	α_k	μ_k^A	α_k^A	μ_k^B	α_k^B
0	0.0625	0.0624	8.7956	0.0458	9.1069	0.5927	96.746
1	0.1875	0.1843	8.0008	0.1395	8.7765	0.5953	31.915
2	0.3125	0.2983	7.2786	0.2354	8.3306	0.6004	18.734
3	0.4375	0.4008	6.6024	0.3327	7.7598	0.6082	12.909
4	0.5625	0.4903	5.9467	0.4301	7.0452	0.6187	9.5003
5	0.6875	0.5665	5.2800	0.5255	6.1489	0.6324	7.1317
6	0.8125	0.6306	4.5443	0.6152	4.9767	0.6502	5.1980
7	0.9375	0.6839	3.5354	0.6918	3.0569	0.6757	2.9853

12 Interferometric phases of the subsystems, and gauge invariance

The Uhlmann holonomy of Sec. 6 is not the only geometric phase a mixed state carries, and it is not the one an interferometer reads. This section derives the *interferometric* (Sjöqvist) phase [9] of both reduced states under the same global cycle, establishes its gauge invariance, and shows in what precise sense it is complementary to the Uhlmann phase.

12.1 Definition and parallel transport

Let $\rho(\phi) = U(\phi)\rho_0 U^\dagger(\phi)$ with $\rho_0 = \sum_k \lambda_k |k\rangle\langle k|$. The interferometric phase is the Pancharatnam phase of the fringe pattern produced when U is applied in one arm of a Mach–Zehnder interferometer, with the dynamical phase of every eigenchannel removed:

$$\Phi^{\text{int}} = \operatorname{Arg} \sum_k \lambda_k \langle k| U(\tau) |k\rangle \exp\left[-\int_0^\tau \langle k| U^\dagger \dot{U} |k\rangle \, d\phi \right]. \tag{55}$$

Equivalently, one asks for the unitary $U_{\rm par} = U\,X$ that generates the same path of density matrices, $[X, \rho_0] = 0$, $X(0) = \mathbb{1}$, and satisfies the N parallel-transport conditions

$$\langle k|\, U_{\rm par}^\dagger \dot{U}_{\rm par}\, |k\rangle = 0, \qquad k = 1, \dots, N, \tag{56}$$

whereupon $\Phi^{\rm int} = \operatorname{Arg}\operatorname{Tr}[\rho_0 U_{\rm par}(\tau)]$. For a time-independent generator, $\dot{U} = iGU$, condition (56) is solved by $X = e^{-iG_d\phi}$ with $G_d \equiv \sum_k P_k G P_k$ the part of G that is diagonal in the eigenbasis of ρ_0, $P_k = |k\rangle\langle k|$, so that

$$\boxed{\Phi^{\rm int} = \operatorname{Arg}\operatorname{Tr}\big[\rho_0\, e^{iG\tau} e^{-iG_d\tau}\big]\,.} \tag{57}$$

12.2 Gauge invariance

Two gauge freedoms act here, and the phase must be blind to both.

(G1) Eigenvector phases. A rephasing $|k\rangle \to e^{i\zeta_k}|k\rangle$ leaves every projector P_k — and hence G_d, ρ_0 and Eq. (57) — unchanged. In the form (55) the factor $e^{i\zeta_k}$ cancels between the bra and the ket of each term. Invariance is exact and term by term.

(G2) Which unitary implements the path. The path $\phi \mapsto \rho(\phi)$ fixes U only up to

$$U(\phi) \;\longrightarrow\; U(\phi)\, e^{i\Xi(\phi)}, \qquad [\Xi(\phi), \rho_0] = 0, \qquad \Xi(0) = 0, \tag{58}$$

i.e. up to the stabilizer of ρ_0. Writing $\Xi = \sum_k \xi_k(\phi) P_k$, the endpoint factor in (55) becomes $\langle k|\, U e^{i\Xi}\, |k\rangle = e^{i\xi_k(\tau)}\langle k|\, U\, |k\rangle$, while

$$\langle k|\, \big(Ue^{i\Xi}\big)^\dagger \frac{d}{d\phi}\big(Ue^{i\Xi}\big)\, |k\rangle = \langle k|\, U^\dagger \dot{U}\, |k\rangle + i\,\dot{\xi}_k \quad\Longrightarrow\quad e^{-\int} \;\longrightarrow\; e^{-i\xi_k(\tau)} e^{-\int}. \tag{59}$$

The two factors cancel in *each* term of the sum, so $\Phi^{\rm int}$ is invariant. By contrast the bare Pancharatnam phase $\operatorname{Arg}\operatorname{Tr}[\rho_0 U(\tau)]$ transforms; Fig. 19(a) shows it sweeping the whole circle along a one-parameter family $\Xi_t = t\,\Xi$ while $\Phi^{\rm int}$ stays pinned to the closed form to the discretization error, which vanishes as $(\delta\phi)^2$, Fig. 19(b).

A global phase $U \to e^{i\chi(\phi)}U$ is the special case $\Xi = \chi\mathbb{1}$ of (58); it therefore drops out as well. This will matter below: it is what makes the photon-addition offset $e^{-im\phi}$ common to both branches irrelevant.

12.3 Laboratory frame: the interferometric phase is a weighted sum of eigenstate Berry phases

In the laboratory the cycle is implemented by a phase shifter, $U_s(\phi) = e^{-i\hat{n}_s\phi}$, and two features make (55) collapse: the spectrum of $\hat{n}$ is integral, so $U_s(2\pi) = \mathbb{1}$; and $[U_s, \hat{n}_s] = 0$, so $\nu_k \equiv \langle k|\, \hat{n}_s\, |k\rangle$ is constant along the loop. Hence $\langle k|\, U^\dagger \dot{U}\, |k\rangle = -i\nu_k$ and

$$\boxed{Z_s^{\rm int} = \sum_k \lambda_k\, e^{i\gamma_k}, \qquad \gamma_k = 2\pi\nu_k = 2\pi\,\langle k|\, \hat{n}_s\, |k\rangle\,, \qquad \Phi_s^{\rm int} = \operatorname{Arg} Z_s^{\rm int}} \tag{60}$$

where γ_k is exactly the Berry phase that the *eigenstate* $|k\rangle$ of $\rho_s(0)$ acquires under the same cycle. The interferometric phase of a subsystem is therefore the Pancharatnam sum of the Berry phases of its Schmidt modes, weighted by their populations — the natural mixed-state successor of Sec. 4, and it reduces to the Berry phase when ρ_s is pure.

Both reduced states are rank two with $\lambda_\pm = \frac{1}{2}(1 \pm r)$, $r = \sqrt{1-C^2}$, the same r for A and B by Schmidt isospectrality. Writing $\nu_\pm = \bar{\nu} \pm \delta_s$, Eq. (60) becomes

$$\boxed{Z_s^{\rm int} = e^{i\pi\,{\rm Tr}\,\mathcal{N}_s}\Big[\cos(2\pi\delta_s) + i\,r\,\sin(2\pi\delta_s)\Big]} \tag{61}$$

with $\mathcal{N}_s \equiv \Pi_s \hat{n}_s \Pi_s$ the number operator projected on the two-dimensional support of ρ_s, $\bar{\nu} = \frac{1}{2}\,{\rm Tr}\,\mathcal{N}_s$, and

$$\delta_s = \frac{1}{r}\left[\frac{(1-2p_s)}{2}\big(\mathcal{N}_{00}^s - \mathcal{N}_{11}^s\big) + 2c_0^s\mathcal{N}_{01}^s\right], \qquad c_0^s = \sqrt{p_s(1-p_s) - \big(\tfrac{C}{2}\big)^2}. \tag{62}$$

In the Gram–Schmidt basis (6) of a branch pair $\{|\alpha, m\rangle, |\alpha, n\rangle\}$ the matrix $\mathcal{N}$ is built from quantities already in hand,

$$\mathcal{N}_{00} = \bar{n}_{\alpha,m}, \qquad \mathcal{N}_{01} = \frac{\kappa_\alpha - g_\alpha \bar{n}_{\alpha,m}}{N_1}, \qquad \mathcal{N}_{11} = \frac{\bar{n}_{\alpha,n} - 2g_\alpha\kappa_\alpha + g_\alpha^2\bar{n}_{\alpha,m}}{N_1^2}, \tag{63}$$

$N_1 = \sqrt{1-g_\alpha^2}$, with $\kappa_\alpha = g_{mn}Q_{mn}$ of Eq. (78). Two structural remarks follow. First, the offset $\pi\,{\rm Tr}\,\mathcal{N}_s$ is a property of the logical subspace alone: it does not depend on μ, hence carries no entanglement information. Second, all the entanglement dependence sits in the bracket of Eq. (61), through $r = \sqrt{1-C^2}$ and through δ_s.

12.4 Visibility, and a single-mode witness of the concurrence

The modulus of (61) is the fringe contrast, and it is a direct function of the concurrence:

$$\boxed{\mathcal{V}_s = \big|Z_s^{\rm int}\big| = \sqrt{1 - C^2\sin^2(2\pi\delta_s)}} \qquad \Longleftrightarrow \qquad \mathcal{V}_s^2 + C^2\sin^2(2\pi\delta_s) = 1. \tag{64}$$

Since δ_s is fixed by the local parameters through Eqs. (62)–(63), the relation inverts:

$$\boxed{C = \frac{\sqrt{1-\mathcal{V}_s^2}}{|\sin(2\pi\delta_s)|}} \tag{65}$$

The interference contrast of one mode alone determines the concurrence of the global entangled PACS. Table 5 evaluates Eq. (65) on sixteen configurations of both families and both subsystems; it reproduces the exact concurrence $C = 2\mu\nu N_1 N_2/\mathcal{N}^2$ of Eq. (19) to 3×10^{-14} in every case. Note that $\mathcal{V}_s = 1$ whenever $\sin 2\pi\delta_s = 0$: the witness needs the loop to be tuned away from those blind points, which is what the α-dependence of $\mathcal{N}_s$ provides. Figure 17 shows the phase, the contrast, the reconstructed concurrence, and simulated Mach–Zehnder fringes.

12.5 The same construction on the logical loop, and the comparison with Uhlmann

Equations (60)–(65) refer to the laboratory realization of the cycle. The Uhlmann analysis of Sec. 6 was carried out on the reduced 2×2 loop (21), i.e. in the co-moving Gram–Schmidt frame that follows the coherent amplitude $\alpha \to \alpha e^{-i\phi}$ and returns to itself at $\phi = 2\pi$. The two descriptions are related by a closed loop of frames and are not term-by-term equal — a reduced-state geometric phase depends on which unitary realizes the cycle, not merely on the reduced state itself. To compare like with like we also evaluate (57) on the logical loop, where $G = \frac{\omega_s}{2}\sigma_z$:

$$\boxed{Z_s^{\mathrm{int}}\big|_{\log} = (-1)^{\omega_s}\Big[\cos\gamma_s - i\,r\sin\gamma_s\Big], \qquad \gamma_s = \pi\omega_s\,\frac{1-2p_s}{r} = \pi\omega_s\cos\theta_s} \tag{66}$$

with $\cos\theta_s = (1-2p_s)/r$ the tilt of the reduced Bloch vector already used in Sec. 7. Comparing with Eq. (30), the two mixed-state phases differ by exactly one thing — how parallel transport treats the generator in the eigenbasis of ρ:

$$\text{Uhlmann:} \quad G \to G_d + C\,G_{\mathrm{od}}, \qquad\qquad \text{Sjöqvist:} \quad G \to G_{\mathrm{od}}. \tag{67}$$

Uhlmann keeps the diagonal part intact and rescales the off-diagonal part by the concurrence, Eq. (29); the interferometric prescription deletes the diagonal part and leaves the off-diagonal part alone. This suggests the one-parameter interpolation $G_{\mathrm{eff}}(s) = G_d + s\,G_{\mathrm{od}}$, for which the holonomy amplitude is

$$Z(s) = (-1)^{\omega}\left[\cos(\pi b_s) - i\,\frac{\omega(1-2p)}{b_s}\sin(\pi b_s)\right], \qquad b_s = \omega\sqrt{\frac{(1-2p)^2 + s^2\big(4p(1-p) - C^2\big)}{1-C^2}}, \tag{68}$$

which contains all three cases:

s	b_s	phase
$s=1$	ω	no transport: $Z=1$, $\Phi=0$
$s=C$	$\omega\sqrt{1+C^2-4p(1-p)}$	Uhlmann, Eq. (30)
$s=0$	$\omega(1-2p)/r$	interferometric, Eq. (66)

On the singular locus $p=1/2$ the interpolation collapses to

$$\boxed{b_s\big|_{p=1/2} = \omega\,s} \tag{69}$$

independently of C. The nodal condition $|b| = k + \frac{1}{2}$ therefore selects the *horizontal* lines $s = (2k+1)/2\Delta$ of the (C,s) plane, Fig. 18(b). The Uhlmann phase lives on the diagonal $s=C$ and crosses each of those Δ lines exactly once, at $C_k = (2k+1)/2\Delta$ — which is Eq. (33) read geometrically. The interferometric phase lives on the axis $s=0$ and crosses none of them.

12.6 No interior singularities, and the one that survives

Directly from (66),

$$\big|Z_s^{\mathrm{int}}\big|_{\log} = \sqrt{1 - C^2\sin^2\gamma_s} \;\geq\; \sqrt{1-C^2} = r > 0 \qquad \text{for } C<1, \tag{70}$$

the same bound as in the laboratory frame. *The interferometric phase of either subsystem has no singularity anywhere in the interior of the physical region*, in sharp contrast with the Δ Uhlmann nodes of Eq. (33); Fig. 18(c) contrasts the two moduli over the (p,C) plane. In particular, on the locus $p=1/2$ one has $\gamma_s = 0$ and

$$\Phi_s^{\mathrm{int}}\Big|_{p=1/2} = \pi\Delta \pmod{2\pi} \qquad \text{with} \qquad \mathcal{V}_s = 1, \tag{71}$$

a constant parity with full contrast: exactly where the Uhlmann phase resolves the concurrence most finely — the $\mathbb{Z}_2$ staircase of Eq. (44) — the interferometric phase is completely blind to it, Fig. 18(a).

The single exception is the boundary $C = 1$. Maximal entanglement forces $r = 0$, hence $p = 1/2$ and $\rho_s = \mathbb{1}/2$; the eigenbasis becomes undetermined and the gauge group of Sec. 12.2 grows from the stabilizer $U(1) \times U(1)$ to all of $U(2)$. Equation (57) then depends on which basis one calls the eigenbasis: for a basis tilted by θ_Q,

$$Z^{\text{int}}\big|_{C=1} = (-1)^{\omega} \cos\big(\pi\omega \cos\theta_Q\big), \tag{72}$$

which is real, takes any value in $[-1, 1]$, and vanishes whenever $\omega \cos\theta_Q = k + \frac{1}{2}$. Figure 19(c) verifies this numerically. So the interferometric phase does possess nodes, but they are all pushed onto the maximally entangled boundary and are gauge artefacts of the degeneracy, whereas the Uhlmann nodes sit at interior, gauge-invariant concurrences C_k. The structural reason is that the Uhlmann bundle carries the full $U(2)$ as its structure group everywhere, while the interferometric construction carries only the stabilizer of ρ — a smaller group, and correspondingly a less singular phase. In the PACS model $C = 1$ is reached in the Fock limit $\alpha \to 0$ at $\mu = \nu$, where the two branches become orthogonal.

12.7 Asymmetry between A and B

On the logical loop the crossed family has $p_A = p_B$ and $\omega_B = -\omega_A$, so $\gamma_B = -\gamma_A$ and

$$\Phi_B^{\text{int, log}} = -\Phi_A^{\text{int, log}} \tag{73}$$

exactly, mirroring the Uhlmann antisymmetry (49); in the branch-correlated family $\omega_A = \omega_B$ and the asymmetry is carried entirely by $p_A \neq p_B$, i.e. by $g_\alpha \neq h_\beta$ — the same mechanism as in Sec. 10. In the laboratory frame even the crossed antisymmetry is broken, but only by the offsets $\pi \operatorname{Tr} \mathcal{N}_A \neq \pi \operatorname{Tr} \mathcal{N}_B$ of Eq. (61), which are properties of the two local subspaces and not of the entanglement: the entanglement-dependent bracket still obeys $\delta_B \leftrightarrow \delta_A$ symmetry constraints inherited from the shared r. Table 5 lists both frames side by side.

13 Numerical validation

Every closed form is checked against an independent construction.

Berry sector: truncated two-mode Fock space. Because the global cycle is diagonal in the Fock basis, $e^{-i(\hat{n}_A+\hat{n}_B)\phi} |j\rangle_A |k\rangle_B = e^{-i(j+k)\phi} |j\rangle_A |k\rangle_B$, the evolution can be applied exactly. We build the full coefficient matrix C_{jk} with 90 levels per mode and evaluate overlaps, matrix elements and $\langle \hat{n}_A + \hat{n}_B \rangle$ by direct summation, with no recourse to hypergeometric identities.

Uhlmann sector: two connection-free holonomies. Method 1 is the discrete polar-decomposition algorithm, which implements the parallel-transport condition $W_j^\dagger W_{j+1} \geq 0$ directly: with $\sqrt{\rho_j}\sqrt{\rho_{j+1}} = V_j P_j$ one has $U_{j+1} = V_j^\dagger U_j$ and $\Phi_U = \operatorname{Arg} \operatorname{Tr}[\rho_0 U_{\text{hol}}]$. Method 2 integrates $\dot{U} = -i\Lambda U$ after solving Eq. (26) as a linear system at each step. Neither uses a closed-form connection. Both are compared against the full complex Z_U, not merely $|Z_U|$: it is the modulus alone that is blind to the placement of the scalar $(-1)^\omega$ discussed after Eq. (30), and to the orientation of the loop, so a modulus-only check would not have caught either.

Interferometric sector: the definition itself. Equation (55) is evaluated directly on a discretized family of unitaries — endpoint matrix elements times the numerically integrated transport factor — with no closed form anywhere, and compared to Eq. (66); the residual is $O((\delta\phi)^2)$. Gauge invariance is tested by pushing the same loop through a one-parameter family of gauge transformations (58) built from random Fourier coefficients with $\xi_k(2\pi) \neq 0$. In the laboratory frame, Eq. (60) is obtained by direct diagonalization of the exact truncated-Fock $\rho_s(0)$, so the closed form (61) is never used in the reference value.

Table 2: Independent validation. "Fock" denotes direct summation in the truncated two-mode space; "polar" and "ODE" the two connection-free holonomy algorithms.

Quantity	Checked against	Max. discrepancy
$\langle \alpha, m\vert\beta, n\rangle$, Eq. (76)	Fock	2×10^{-16}
$\kappa_\alpha = g_{mn}Q_{mn}$, Eq. (78)	Fock	6×10^{-15}
Φ_B, Eq. (12)	Fock	2×10^{-14}
Mode split $\Phi_B = \Phi_B^{(A)} + \Phi_B^{(B)}$	Closed forms	9×10^{-16}
Symmetric case $\Phi_B = 2\Phi_B^{(A)}$	Closed forms	0 (exact)
Affine law, Eq. (14)	Direct evaluation	10^{-10}
p_s, C, ω_s, Eqs. (9)–(10)	Fock + winding fit	10^{-9}
Z_U (complex), Eq. (30)	Polar holonomy	3×10^{-5}
Z_U (complex), Eq. (30)	Transport ODE	3×10^{-5}
Tilt identity, Eq. (48)	Eq. (30)	2×10^{-16}
Nodes C_k, Eq. (33)	$\vert Z_U\vert$ at the node	10^{-14}
Antisymmetry, Eq. (49)	Closed form	0 (exact)
$C = 2\mu\nu N_1 N_2/\mathcal{N}^2$, Eq. (19)	$2\vert\det M\vert$ and Fock	10^{-16}
$\sqrt{p_A p_B} = C/2$, Eq. (39)	Closed forms	10^{-16}
$p = C/2$ and Eq. (43) (balanced sym.)	Eq. (30)	2×10^{-16}
$\mathbb{Z}_2$ reduction on $p = 1/2$, Eq. (44)	$\operatorname{Im} Z_U$	0 (exact)
Stable overlap (53) (Kummer)	Fock, up to $\alpha = 60$	4×10^{-15}
$\Delta = 8$ nodes, Table 1	p, b, $\vert Z_U\vert$ at each node	5×10^{-14}
Co-moving frame, Eqs. (23)–(24)	Exact Fock $\rho_s(\phi)$, 2×2 block	9×10^{-16}
Z_s^{int} lab, Eq. (60)	Eq. (61)	4×10^{-14}
Witness (65)	Exact C, Eq. (19)	3×10^{-14}
Z_s^{int} log, Eq. (66)	Definition (55), discretized	3×10^{-6}
Gauge invariance (58)	Spread over gauges, $\delta\phi = 2\pi/8000$	4×10^{-5}
$Z(s{=}C) = Z_U$, $Z(s{=}0) = Z^{\text{int}}$, Eq. (68)	Eqs. (30), (66)	7×10^{-15}
Degenerate limit, Eq. (72)	Eq. (57), direct	3×10^{-15}

As a sharp discriminant, the nodes predicted by Eq. (33) give $|\operatorname{Tr}[\rho_0 U_{\text{hol}}]| \sim 10^{-15}$ in the ODE holonomy, i.e. genuine zeros, whereas the values $C = 2\Delta/(2k+1)$ that follow from the sandwich connection give $|\operatorname{Tr}[\rho_0 U_{\text{hol}}]| = 0.31, 0.90, 0.94$ for $\Delta = 2$, $k = 2, 3, 4$ — not zeros at all.

14 Results

All figures use real amplitudes; "balanced" means $\mu = \nu = 1/\sqrt{2}$ and "symmetric" means $\beta = \alpha$.

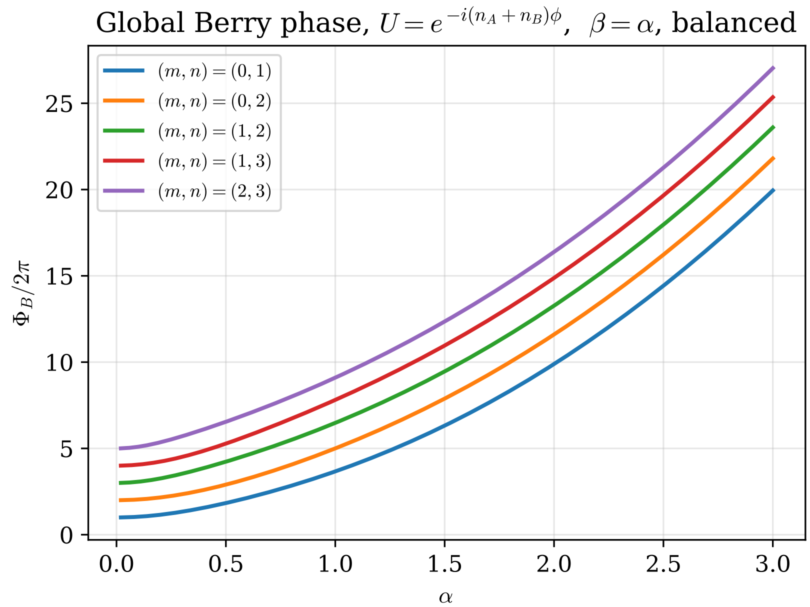

Figure 1: Global Berry phase $\Phi_B/2\pi = \langle \hat{n}_A + \hat{n}_B \rangle$, Eq. (12), versus α for several photon-addition pairs in the balanced symmetric state. Growth is smooth because the connection is constant along the cycle; changing (m, n) shifts the curve through both the diagonal mean-photon terms and the interference term of Eq. (14).

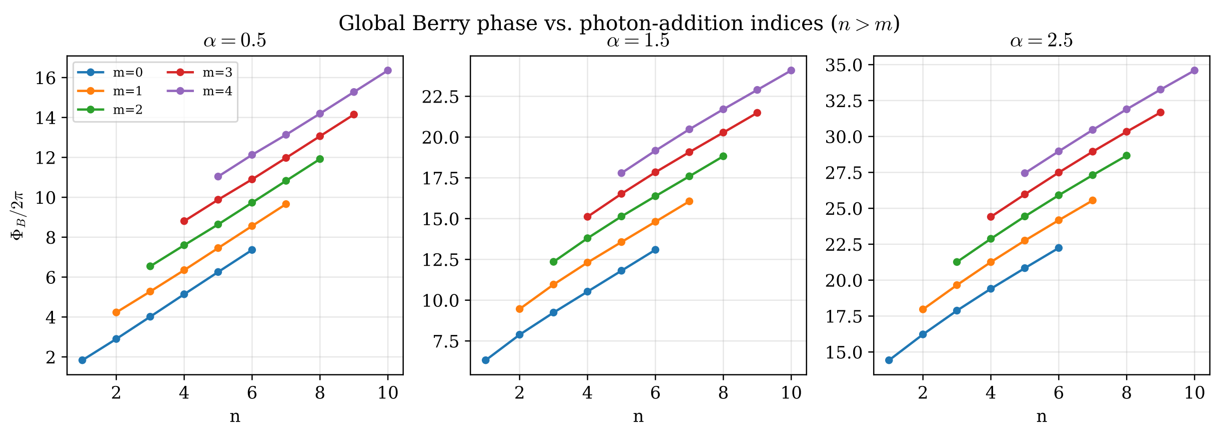

Figure 2: Discrete dependence of the global Berry phase on the photon-addition indices at three coherent amplitudes. Only $n > m$ is shown, the ordering in which the overlap formula (76) is written.

Table 3: Critical amplitudes $\alpha_g(m, n)$ from $|g_{mn}(\alpha_g)|^2 = 1/2$.

$n \backslash m$	0	1	2	3	4
1	1.0000				
2	2.1094	0.7221			
3	3.2756	1.8000	0.5921		
4	4.4576	3.0354	1.5829	0.5135	
5	5.6464	4.2685	2.8269	1.4235	0.4596

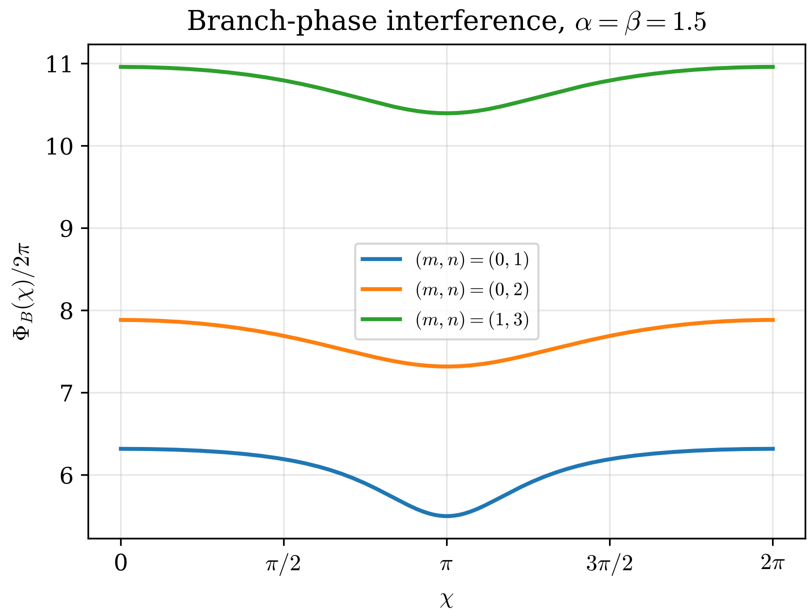

Figure 3: Interference fringes produced by the tunable relative branch phase χ. The ϕ-cycle itself shows no oscillation because its connection, though containing the coherent cross term, is ϕ-independent; χ makes that interference explicit.

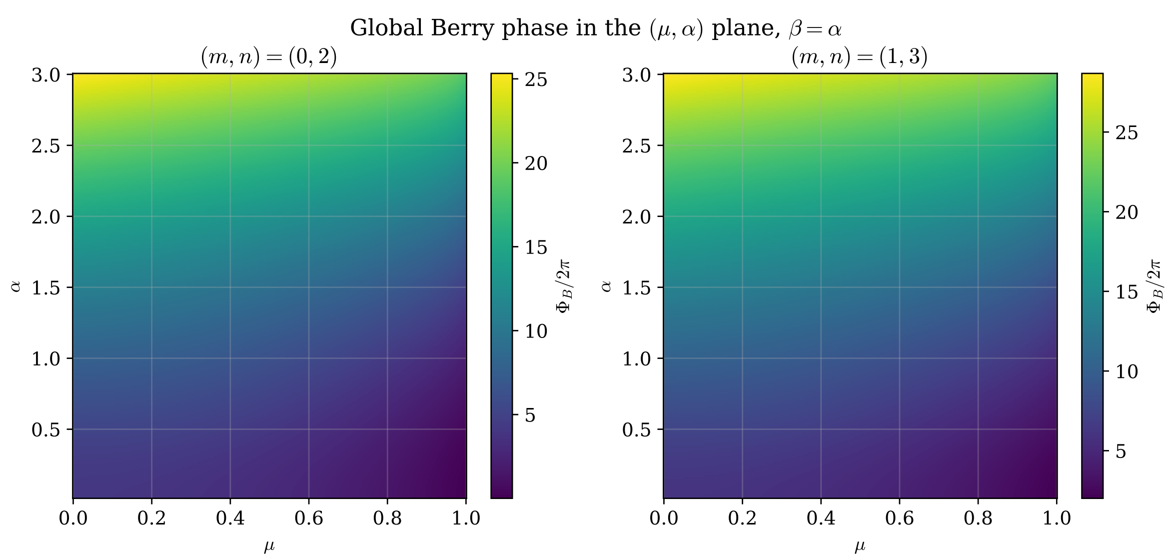

Figure 4: Global Berry phase in the (μ, α) plane. The map is completely smooth: a pure-state Berry phase never develops branch cuts, in contrast with the Uhlmann maps below.

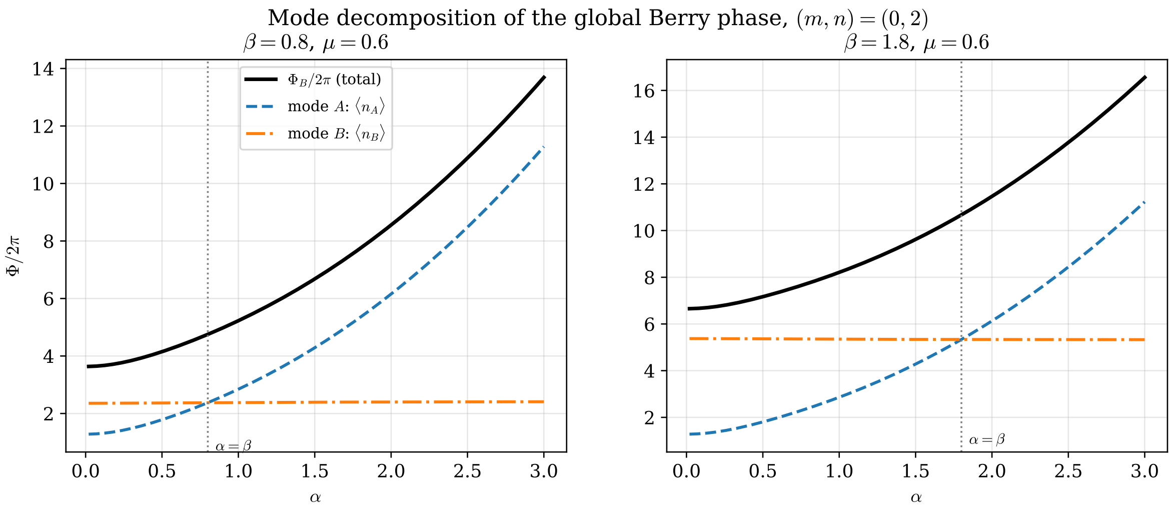

Figure 5: Exact mode decomposition (13) of the global Berry phase for two fixed values of β. The total (black) is the sum of the two single-mode contributions; the mode whose amplitude is held fixed contributes an almost constant amount, so the split is markedly unequal away from $\alpha = \beta$.

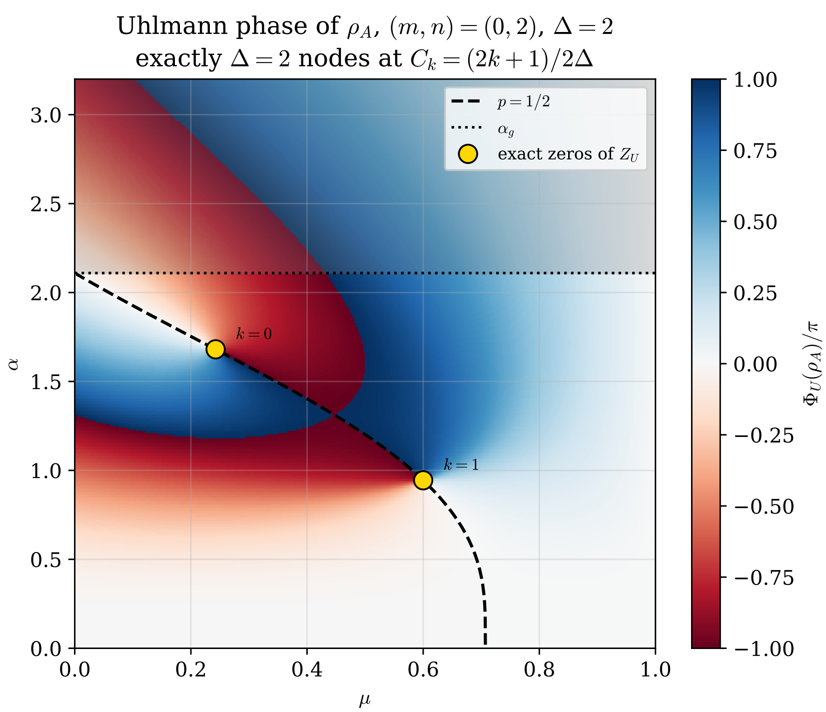

Figure 6: Principal Uhlmann phase of ρ_A in the (μ, α) plane, Eq. (30), for the symmetric family. The dashed curve is the analytic condition $p = 1/2$, Eq. (34); the dotted line is α_g and the shaded region above it cannot contain nodes. Gold circles are the exact zeros of Z_U: for $\Delta = 2$ there are exactly two, labelled $k = 0, 1$, at $C_k = (2k + 1)/2\Delta$.

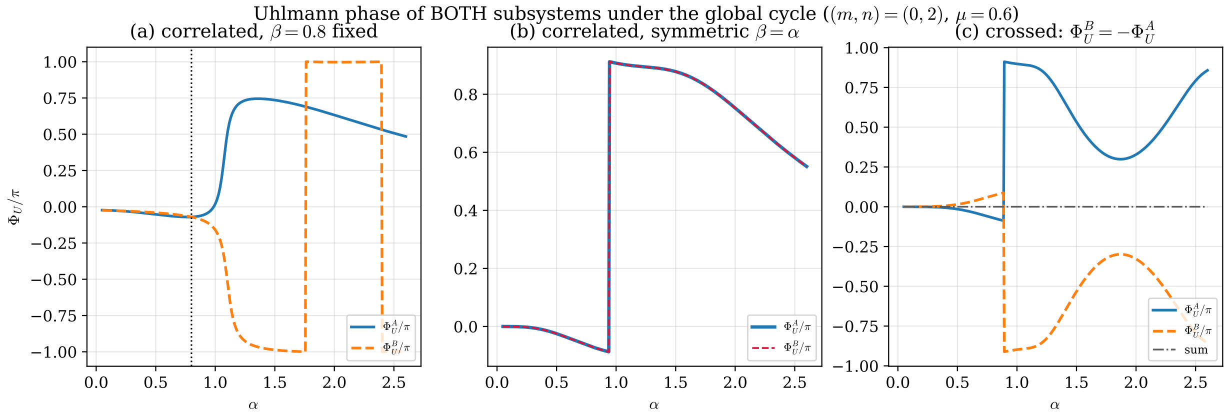

Figure 7: Uhlmann phase of *both* subsystems under the global cycle. (a) Branch-correlated family with $\beta = 0.8$ fixed: the two phases are different curves, meeting only at $\alpha = \beta$, even though they share a concurrence. (b) Symmetric family $\beta = \alpha$: $p_A = p_B$ by Eq. (9) and the curves coincide exactly. (c) Crossed family, where $\omega_B = -\omega_A$: the exact antisymmetry (49) holds and the sum (dash-dotted) vanishes identically.

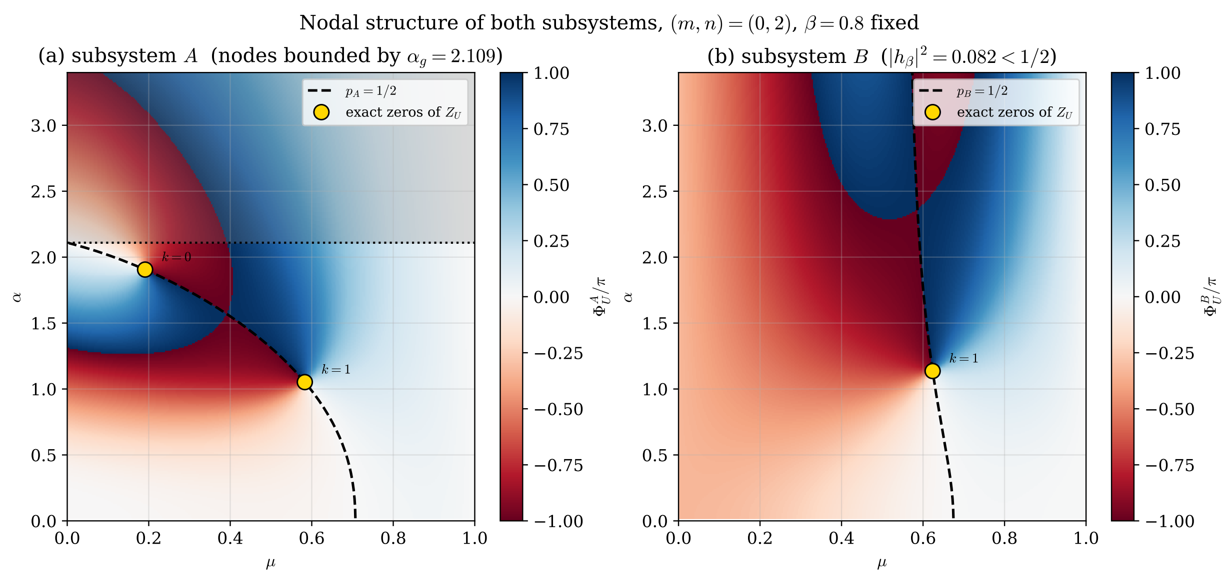

Figure 8: Nodal structure of both subsystems in the (μ, α) plane at fixed $\beta = 0.8$. Both share the same nodal concurrences C_k, Eq. (33), but not their coordinates: the $p_A = 1/2$ curve bends and keeps the A nodes below α_g, whereas the existence criterion for B depends on β alone, so its $p_B = 1/2$ curve is nearly vertical and its nodes are not bounded by α_g (the $k = 0$ node of B lies at $\alpha \approx 5.4$, outside the plotted range).

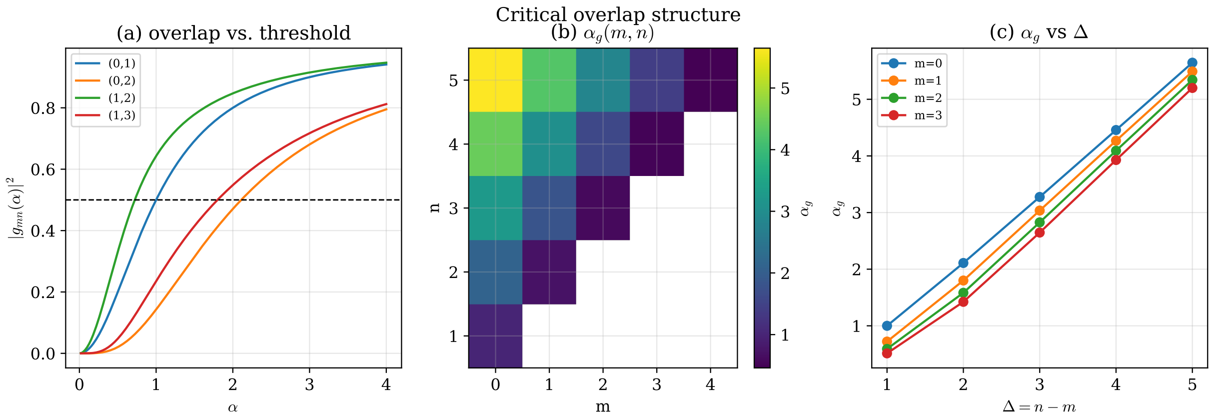

Figure 9: Critical overlap structure. (a) $|g_{mn}(\alpha)|^2$ against the threshold 1/2. (b) The discrete critical function $\alpha_g(m,n)$ over the grid of Table 3. (c) α_g versus Δ for several fixed m: larger photon-addition separation extends the nodal region to larger coherent amplitude.

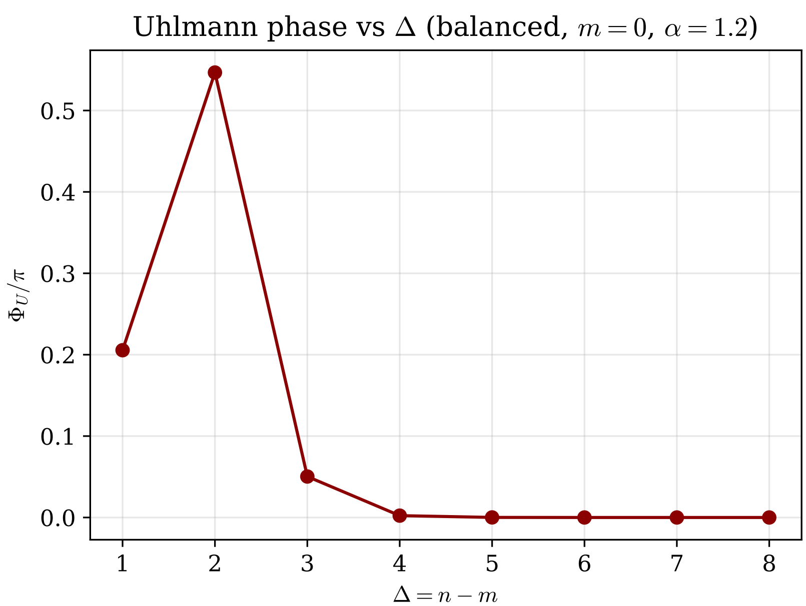

Figure 10: Principal Uhlmann phase versus photon-addition separation Δ for the balanced symmetric state. The nonmonotonic behaviour reflects the dependence of $b = \Delta C$ on Δ through the overlap, combined with principal-branch wrapping of $\mathrm{Arg}(\cdot) \in (-\pi, \pi]$.

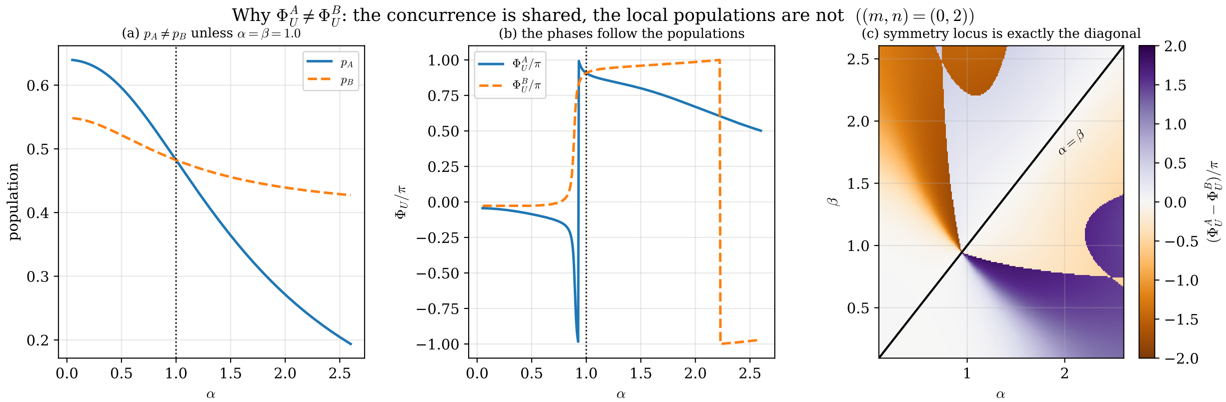

Figure 11: Origin of the $A \leftrightarrow B$ asymmetry (Sec. 10). (a) The local populations differ, crossing only at $\alpha = \beta$. (b) The two Uhlmann phases track those populations. (c) Map of $\Phi_U^A - \Phi_U^B$ in the (α, β) plane: the symmetry locus is exactly the diagonal $\alpha = \beta$, where the two local overlaps coincide.

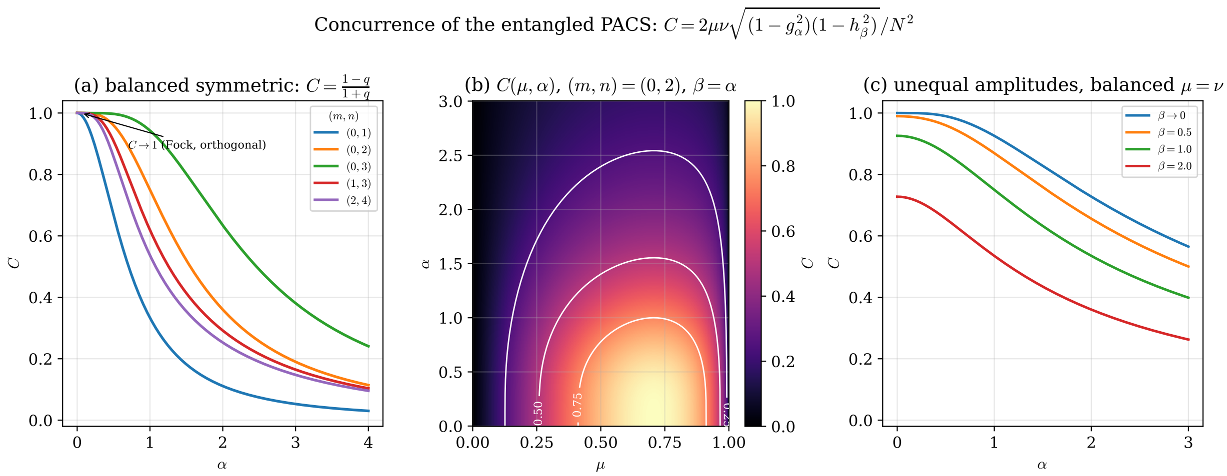

Figure 12: Concurrence of the entangled PACS, Eq. (19). (a) Balanced symmetric family: $C = (1-q)/(1+q)$ decreases monotonically from 1 at $\alpha \to 0$ (where the PACS reduce to orthogonal Fock states) towards 0 at large α (parallel branches). Larger Δ keeps the branches distinguishable longer. (b) C over the (μ, α) plane with contours. (c) Unequal amplitudes: fixing β caps the attainable concurrence, since C carries the factor $\sqrt{(1-g_\alpha^2)(1-h_\beta^2)}$.

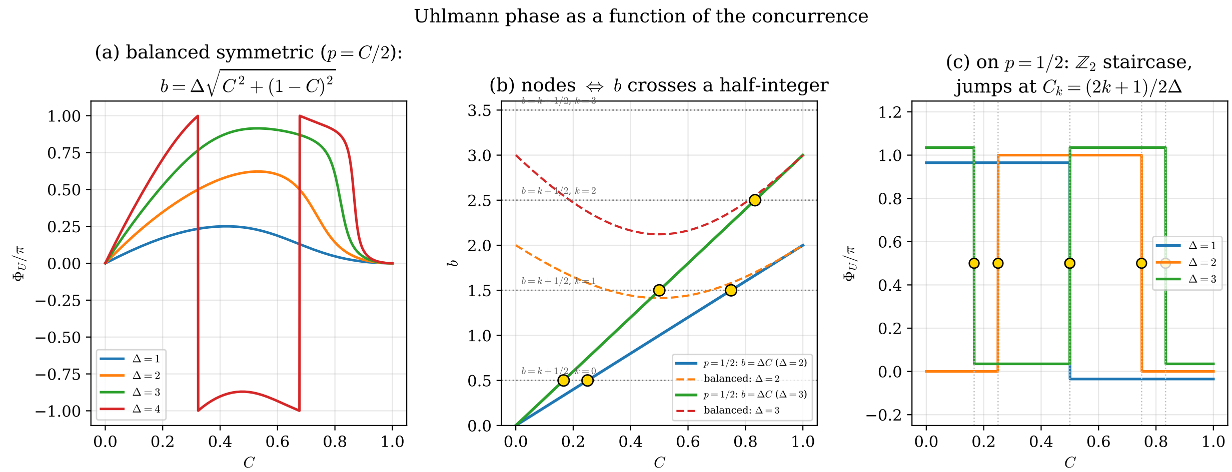

Figure 13: Uhlmann phase as a function of the concurrence. (a) In the balanced symmetric family $p = C/2$, so Φ_U is a pure function of C through $b = \Delta\sqrt{C^2 + (1-C)^2}$, Eq. (43); it returns to 0 at both $C = 0$ and $C = 1$. (b) b versus C on the two loci, with the half-integer levels dotted: on $p = 1/2$ the line $b = \Delta C$ crosses them exactly at the nodal concurrences (gold). The balanced curve may also cross a half-integer, but those are not nodes because $p \neq 1/2$ leaves $\operatorname{Im} Z_U \neq 0$. (c) On $p = 1/2$ the phase is the $\mathbb{Z}_2$ staircase of Eq. (44), flipping between 0 and π exactly at the Δ nodal concurrences (curves offset vertically for visibility).

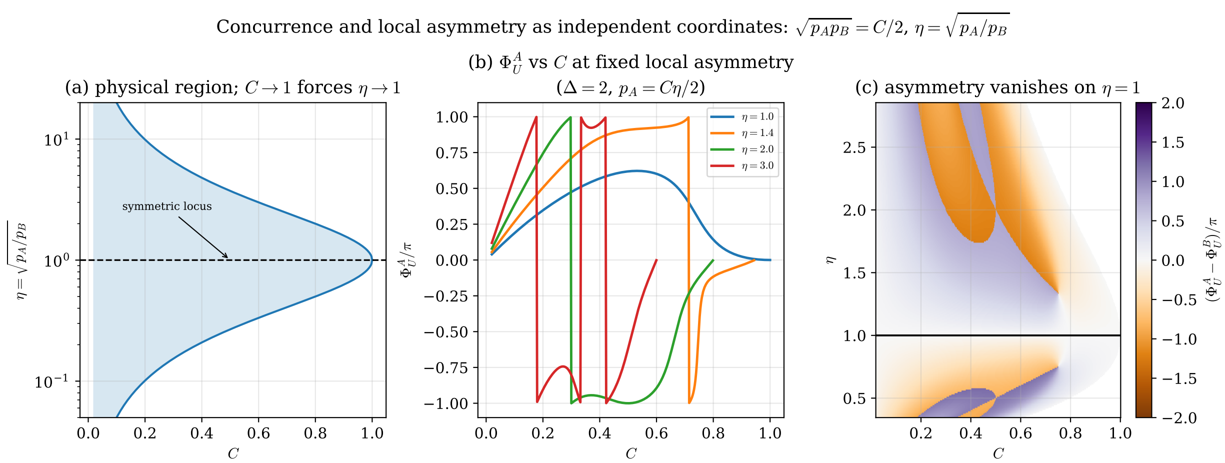

Figure 14: Concurrence and local asymmetry as independent coordinates, Eqs. (39)–(40). (a) The physical region (41) in the (C, η) plane; it is symmetric under $\eta \to 1/\eta$ and pinches to the single point $\eta = 1$ as $C \to 1$, so maximal entanglement enforces $A \leftrightarrow B$ symmetry. (b) Φ_U^A versus C at fixed local asymmetry η. (c) $\Phi_U^A - \Phi_U^B$ over the same plane: the asymmetry vanishes identically on the line $\eta = 1$.

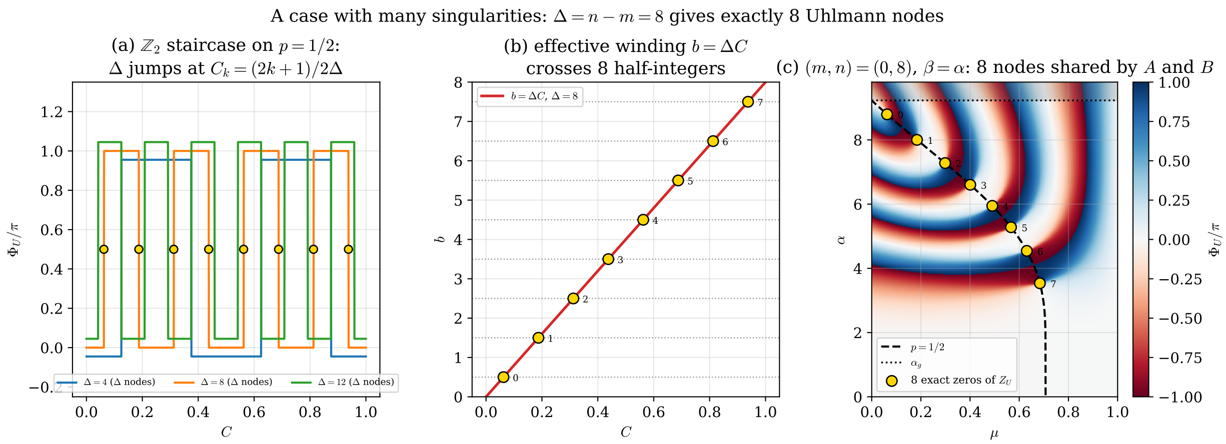

Figure 15: A case with many singularities, $\Delta = n - m = 8$. (a) The $\mathbb{Z}_2$ staircase of Eq. (44) now takes Δ steps; curves for $\Delta = 4, 8, 12$ are offset vertically. (b) The effective winding $b = \Delta C$ crosses $\Delta = 8$ half-integers, one per node (gold, labelled by k). (c) The eight nodes in the (μ, α) plane for the symmetric family $\beta = \alpha$, strung along the $p = 1/2$ curve and all below $\alpha_g = 9.23$; since $p_A = p_B$ there, these eight points are the nodes of *both* subsystems.

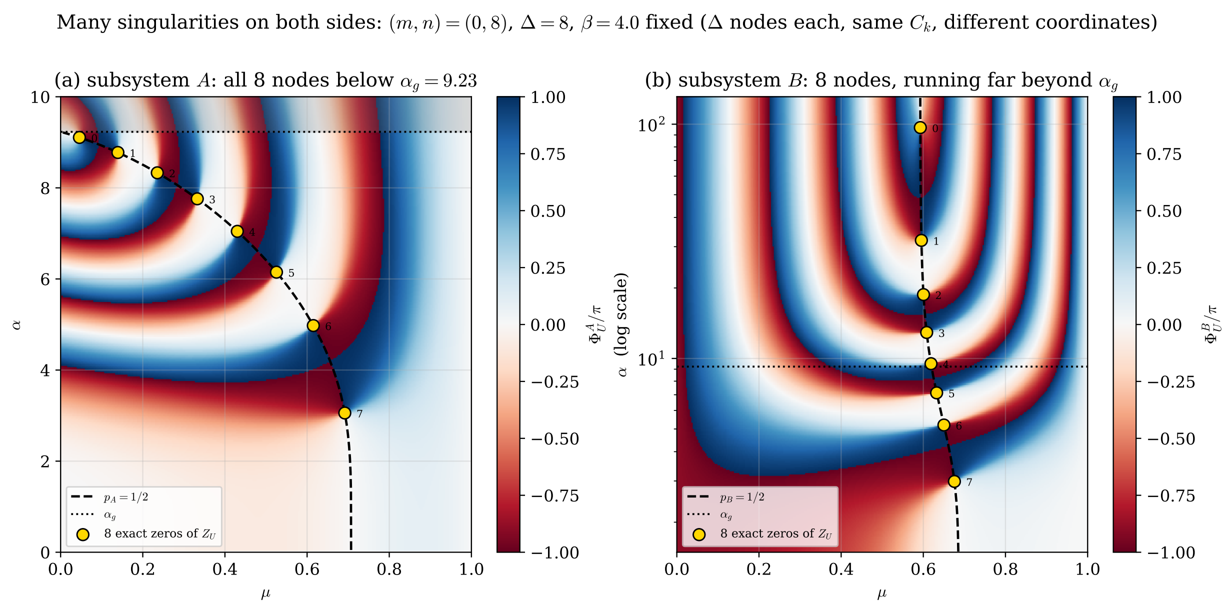

Figure 16: The same $\Delta = 8$ case with $\beta = 4$ fixed, so that the two subsystems separate. Each side still carries exactly eight nodes at the same concurrences C_k, but at wildly different coordinates: (a) the eight nodes of A all lie below $\alpha_g = 9.23$ (shaded region excluded); (b) those of B, bounded only by the criterion (38) on β, run out to $\alpha \approx 97$ — note the logarithmic α axis. Numerical values in Table 1.

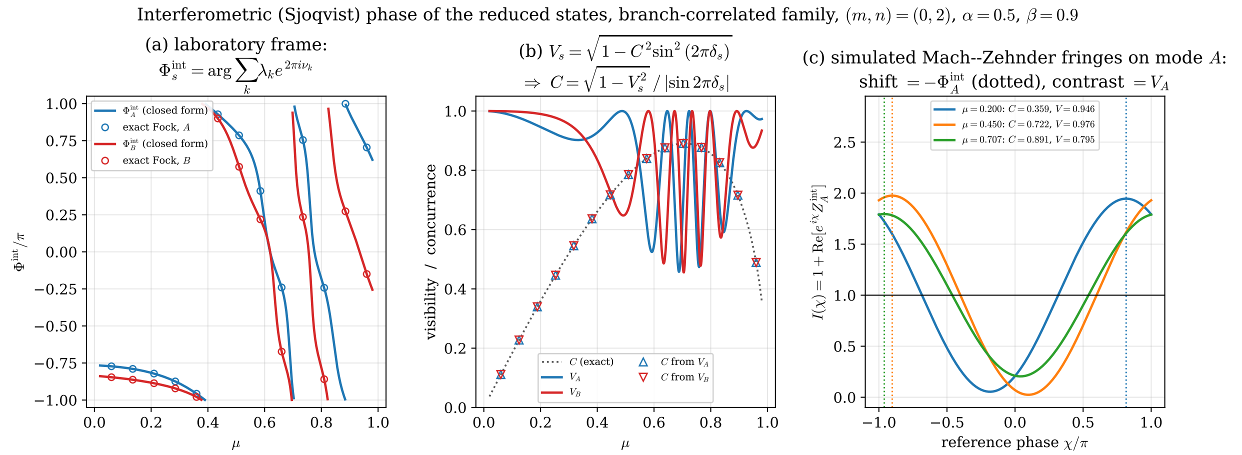

Figure 17: Interferometric (Sjöqvist) phase of the two reduced states in the laboratory frame, Eqs. (60)–(61), for the branch-correlated family with $(m,n) = (0,2)$, $\alpha = 0.5$, $\beta = 0.9$. (a) $\Phi_A^{\rm int}$ and $\Phi_B^{\rm int}$ versus μ: lines are the closed form, markers the exact truncated-Fock evaluation of $\operatorname{Arg}\sum_k \lambda_k e^{2\pi i\nu_k}$ (agreement 4×10^{-14}). The two subsystems differ because the offsets $\pi\,\mathrm{Tr}\,\mathcal{N}_s$ and the splittings δ_s are local properties of each mode. (b) Fringe contrast $\mathcal{V}_s = \sqrt{1 - C^2\sin^2 2\pi\delta_s}$ (solid) against the exact concurrence (dotted); triangles are the concurrence *reconstructed* from the contrast of a single mode through the witness (65), and fall on the exact curve. (c) Simulated Mach–Zehnder fringes $I(\chi) = 1 + \mathrm{Re}[e^{i\chi}Z_A^{\rm int}]$ at three values of μ: the fringe shift is $-\Phi_A^{\rm int}$ (dotted verticals) and the contrast is $\mathcal{V}_A$.

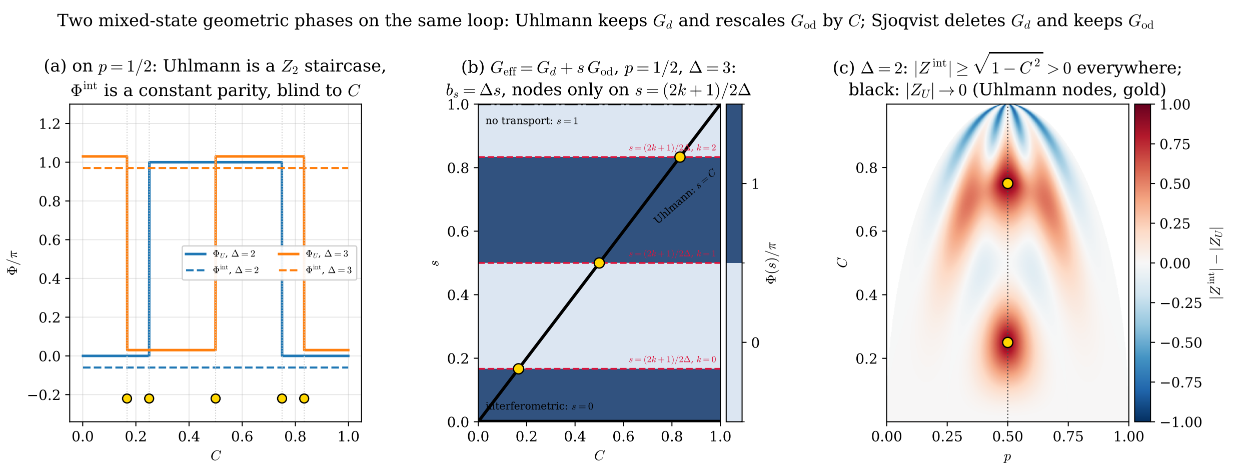

Figure 18: The two mixed-state geometric phases on the same logical loop, Eq. (67). (a) On $p = 1/2$: the Uhlmann phase is the $\mathbb{Z}_2$ staircase of Eq. (44), jumping at the quantized concurrences $C_k = (2k+1)/2\Delta$ (gold), while the interferometric phase is the constant parity $\pi\Delta$ of Eq. (71), blind to C (curves offset vertically for legibility). (b) The interpolating family $G_{\rm eff} = G_d + s\,G_{\rm od}$ of Eq. (68) at $p = 1/2$, $\Delta = 3$: since $b_s = \Delta s$, all nodes lie on the horizontal lines $s = (2k+1)/2\Delta$ (dashed). The Uhlmann phase runs along the diagonal $s = C$ and meets every one of them; the interferometric phase runs along $s = 0$ and meets none; $s = 1$ is the untransported cycle. (c) Difference of moduli over the physical region (p, C) for $\Delta = 2$: $|Z^{\rm int}| \geq \sqrt{1 - C^2} > 0$ everywhere, whereas $|Z_U|$ vanishes at the two gold nodes on $p = 1/2$ (black contour $|Z_U| = 0.02$).

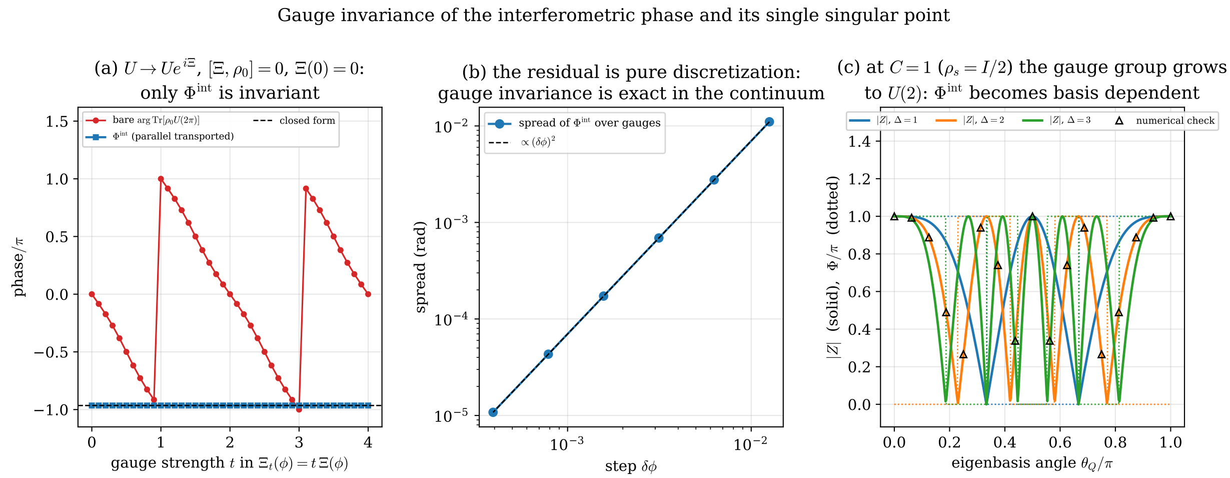

Figure 19: Gauge invariance of the interferometric phase, Sec. 12.2. (a) Under the one-parameter gauge family $U \to U e^{it\Xi(\phi)}$ with $[\Xi, \rho_0] = 0$ and $\Xi(0) = 0$, the bare Pancharatnam phase $\mathrm{Arg}\,\mathrm{Tr}[\rho_0 U(2\pi)]$ sweeps the entire circle while $\Phi^{\rm int}$ stays pinned to the closed form (66) (dashed). (b) The residual spread of $\Phi^{\rm int}$ over gauges scales as $(\delta\phi)^2$: gauge invariance is exact in the continuum and the residual is purely the discretization of the transport integral. (c) At $C = 1$, where $\rho_s = \mathbb{1}/2$ and the gauge group grows to $U(2)$, the phase becomes basis dependent: $|Z^{\rm int}| = |\cos(\pi\Delta\cos\theta_Q)|$ as a function of the eigenbasis angle, Eq. (72), with zeros whenever $\Delta\cos\theta_Q$ is a half-integer (triangles: direct numerical evaluation of Eq. (57)).

Table 4: Complete Uhlmann nodal set, Eqs. (33)–(35). Each Δ supports exactly Δ nodes; all satisfy $p = 1/2$ and $\alpha_k < \alpha_g$. The last column is the numerically evaluated $|Z_U|$.

(m,n)	Δ	k	C_k	μ_k	α_k	α_g	$\lvert Z_U\rvert$
$(0,1)$	1	0	0.5000	0.4472	0.5774	1.0000	4.5×10^{-16}
$(0,2)$	2	0	0.2500	0.2425	1.6807	2.1094	6.1×10^{-17}
		1	0.7500	0.6000	0.9444	2.1094	1.8×10^{-16}
$(0,3)$	3	0	0.1667	0.1644	2.8433	3.2756	3.4×10^{-15}
		1	0.5000	0.4472	2.1094	3.2756	1.8×10^{-16}
		2	0.8333	0.6402	1.3628	3.2756	2.1×10^{-15}
$(1,2)$	1	0	0.5000	0.4472	0.4095	0.7221	5.7×10^{-15}
$(1,3)$	2	0	0.2500	0.2425	1.3763	1.8000	6.1×10^{-17}
		1	0.7500	0.6000	0.7265	1.8000	6.6×10^{-14}
$(1,4)$	3	0	0.1667	0.1644	2.5872	3.0354	5.5×10^{-15}
		1	0.5000	0.4472	1.8378	3.0354	5.0×10^{-15}
		2	0.8333	0.6402	1.1224	3.0354	2.4×10^{-15}
$(2,4)$	2	0	0.2500	0.2425	1.1867	1.5829	3.1×10^{-15}
		1	0.7500	0.6000	0.6130	1.5829	1.8×10^{-16}

Table 5: Interferometric phases of both subsystems. $\Phi_s^{\mathrm{int}}/\pi$ is the laboratory-frame phase (60), evaluated by direct diagonalization of the exact truncated-Fock $\rho_s(0)$; it agrees with the closed form (61) to 4×10^{-14} on every row. $C(\mathcal{V}_s)$ is the concurrence reconstructed from the single-mode fringe contrast through the witness (65); it reproduces the exact C to 3×10^{-14}. The last two columns give the interferometric and Uhlmann phases of the same logical loop, Eqs. (66) and (30), for comparison: note the exact antisymmetry between A and B in the crossed family, and its absence in the laboratory frame.

family	(m,n)	α	β	μ	s	C	δ_s	$\Phi_s^{\mathrm{int}}/\pi$	$\mathcal{V}_s$	$C(\mathcal{V}_s)$	$\Phi_s^{\mathrm{int,log}}/\pi$	$\Phi_U^{\mathrm{log}}/\pi$
corr.	$(0,2)$	0.5	0.9	0.4500	A	0.7223	+1.0489	+0.9021	0.9759	0.7223	-0.5578	-0.1551
					B	0.7223	+1.2923	+0.8577	0.7172	0.7223	-0.9610	-0.2727
corr.	$(0,2)$	0.5	0.9	0.7071	A	0.8906	+0.1190	+0.9604	0.7956	0.8906	-0.1650	+0.0184
					B	0.8906	+0.2305	+0.6539	0.4677	0.8906	-0.7882	+0.0369
corr.	$(0,2)$	1.1	0.8	0.6000	A	0.7480	+0.8099	+0.7985	0.7183	0.7480	-0.1158	+0.4987
					B	0.7480	+0.6992	-0.8057	0.7040	0.7480	+0.1104	-0.5034
corr.	$(1,3)$	1.3	0.7	0.4500	A	0.5257	+1.1058	-0.7164	0.9460	0.5257	-0.2219	+0.9856
					B	0.5257	+0.9245	-0.1837	0.9708	0.5257	+0.5100	-0.9450
corr.	$(2,4)$	0.8	1.6	0.3500	A	0.3373	+1.0136	+0.8698	0.9996	0.3373	+0.2841	+0.8699
					B	0.3373	+1.1967	+0.3783	0.9479	0.3373	-0.6096	-0.9219
cross.	$(0,2)$	1.1	0.8	0.6000	A	0.5574	+0.7078	-0.5238	0.8430	0.5574	-0.6400	+0.8892
					B	0.5574	+0.1424	+0.3322	0.9005	0.5574	+0.6400	-0.8892
cross.	$(0,3)$	1.5	1.2	0.5000	A	0.5972	+1.3190	+0.3278	0.8404	0.5972	-0.8948	+0.9862
					B	0.5972	-0.5687	+0.5484	0.9683	0.5972	+0.8948	-0.9862
cross.	$(0,1)$	0.9	0.9	0.7071	A	0.3817	+0.3345	-0.6999	0.9443	0.3817	+0.3199	+0.2478
					B	0.3817	+0.3345	-0.6999	0.9443	0.3817	-0.3199	-0.2478

15 Discussion

Choosing the global cycle organizes the problem into three clean levels.

Level 1 — the connection is a photon number. Because $\hat{n}_A + \hat{n}_B$ generates the cycle and commutes with itself, the Berry connection is constant and equals the total mean photon number. Everything in the Berry sector is therefore smooth, single-valued and additive across modes, Eq. (13). The interference between branches is present in the connection but invisible in ϕ; it takes a separately controllable branch phase to expose it.

Level 2 — entanglement enters affinely. In the balanced symmetric family the global Berry phase is an affine function of the concurrence, Eq. (14), interpolating between a mean-photon average and a pure interference term. This is as far as a pure-state geometric phase can go: it cannot be singular.

Level 3 — mixed-state holonomy, and it has two sides. Tracing out either mode leaves a genuinely mixed state, and the Uhlmann holonomy can be singular. The compact result (29) — parallel transport multiplies the off-diagonal generator by the concurrence — makes the structure transparent: at $p = 1/2$ the transported generator has magnitude ΔC, and the singularity condition $|b| = k + 1/2$ quantizes C to the finite ladder (33).

The most instructive feature is the interplay between what entanglement fixes and what it does not. The concurrence is shared by the two subsystems and determines *which* values are geometrically singular — the ladder C_k is a property of the pair. But it does not determine *where* those singularities sit, nor even whether the two sides have the same phase: the local populations p_A, p_B, equivalently the tilts θ_A, θ_B of the reduced Bloch vectors relative to each mode's own logical axis, are set by the local nonorthogonality g_α and h_β, which Schmidt's theorem leaves free. Symmetry is recovered exactly on the diagonal $\beta = \alpha$; and in the crossed family, where the local geometries do coincide but the indices are exchanged, one gets not symmetry but the exact antisymmetry $\Phi_U^A + \Phi_U^B = 0$. Entanglement is symmetric; local geometry need not be, and mixed-state holonomy sees both.

Level 4 — what an interferometer actually sees. The Uhlmann holonomy is not measured by letting a mixed state acquire a phase in one arm of an interferometer; the interferometric phase is. Section 12 shows that the two differ by exactly one choice — which part of the generator parallel transport removes, Eq. (67) — and that the choice has sharp consequences. The Uhlmann prescription retains the diagonal generator and damps the off-diagonal one by C, which is what allows the transported winding b to slide continuously with the entanglement and to cross half-integers: the nodes exist *because* C appears inside the generator. The interferometric prescription removes the diagonal generator and leaves the off-diagonal one untouched, so C never enters the effective winding at all; it survives only in the eigenvalue weights, where it can reduce the fringe contrast but can never make it vanish. That is the content of $|Z^{\text{int}}| \geq \sqrt{1 - C^2}$, and it explains why an interferometric measurement of these states is robust where the Uhlmann phase is singular — and, conversely, why the interferometric phase is a poor probe of the nodal structure but an excellent probe of C itself: what it loses in singularity it gains in visibility, Eq. (65). The two are complementary readouts of the same loop, and the interpolation (68) makes the family they belong to explicit.

Finally, the critical amplitude α_g is not a thermodynamic critical point but an overlap-controlled boundary: it is fixed entirely by $|g_{mn}(\alpha_g)|^2 = 1/2$ and separates the region where equal logical-basis population is kinematically accessible from the region where it is not. Larger α makes the two photon-added branches more distinguishable, pushing the achievable population imbalance away from the symmetric point that an exact node requires.

16 Conclusions

Rebuilding the analysis of entangled PACS around the global two-mode cycle $U(\phi) = e^{-i(\hat{n}_A+\hat{n}_B)\phi}$ yields:

(i) a closed-form global Berry phase equal to the total mean photon number, Eq. (12), exactly additive across modes and affine in the concurrence in the balanced symmetric family;

(ii) the Uhlmann connection from the anticommutator transport condition, giving $(G+\Lambda)_{\mathrm{od}} = C\,G_{\mathrm{od}}$ and $b = \omega\sqrt{1+C^2-4p(1-p)}$, validated against two connection-free holonomy algorithms;

(iii) a finite nodal ladder,

$$p = \frac{1}{2}, \qquad C_k = \frac{2k+1}{2\Delta}, \qquad |g_{mn}(\alpha_k)|^2 = \frac{1-C_k}{2}, \qquad k = 0, \ldots, \Delta-1, \tag{74}$$

i.e. exactly Δ isolated Uhlmann singularities per family, common to both subsystems and all bounded by α_g;

(iv) a precise account of the $A \leftrightarrow B$ asymmetry: shared concurrence, unshared local populations, with symmetry on $\beta = \alpha$ and exact antisymmetry $\Phi_U^B = -\Phi_U^A$ in the crossed family;

(v) the interferometric phase of both subsystems in closed form, together with its gauge invariance: in the laboratory frame $\Phi_s^{\mathrm{int}} = \operatorname{Arg}\sum_k \lambda_k e^{2\pi i \nu_k}$ is the population-weighted Pancharatnam sum of the Berry phases of the Schmidt modes, invariant term by term under $|k\rangle \to e^{i\zeta_k}|k\rangle$ and under $U \to U e^{i\Xi}$ with $[\Xi, \rho_0] = 0$;

(vi) an operational consequence: the fringe contrast $\mathcal{V}_s = \sqrt{1 - C^2 \sin^2 2\pi\delta_s}$ of a *single* mode determines the concurrence of the global entangled PACS through Eq. (65);

(vii) a clean separation of the two mixed-state phases, Eq. (67), embedded in the one-parameter family $G_{\mathrm{eff}} = G_d + s\,G_{\mathrm{od}}$ with $b_s = \omega s$ on $p = 1/2$: the Uhlmann phase ($s = C$) crosses all Δ nodal lines, the interferometric phase ($s = 0$) crosses none, and the latter's only singularity is the maximally entangled point where the gauge group grows from $U(1) \times U(1)$ to $U(2)$.

Taken together these results provide an analytic bridge between photon addition, nonorthogonality, bipartite entanglement and mixed-state quantum holonomy, in a model where every step can be written — and independently checked — in closed form.

A Scalar products of photon-added coherent states

Inserting Eq. (3) into $\langle \alpha, m | \beta, n\rangle$ and using Fock orthogonality, only terms with a common level survive. Writing $\ell = n + p$ and $\Delta = n - m \geq 0$,

$$\langle \alpha, m | \beta, n\rangle = A_m A_n\, \alpha^\Delta \sum_{p=0}^{\infty} (\alpha\beta)^p \,\frac{(n+p)!}{(\Delta+p)!\,p!}, \qquad A_n = \frac{e^{-\alpha^2/2}}{\sqrt{n! L_n(-\alpha^2)}}. \tag{75}$$

The key step is to recognize the factorial ratio as a ratio of Pochhammer symbols, $\frac{(n+p)!}{(\Delta+p)!} = \frac{n!}{\Delta!}\frac{(n+1)_p}{(\Delta+1)_p}$, which converts the sum into a confluent hypergeometric series and gives

$$\boxed{\langle \alpha, m | \beta, n\rangle = \frac{e^{-\frac{1}{2}(\alpha^2+\beta^2)}\,\alpha^\Delta\, n!}{\Delta!\,\sqrt{m!\,n!\,L_m(-\alpha^2) L_n(-\beta^2)}}\;{}_1F_1(n+1; \Delta+1; \alpha\beta)} \tag{76}$$

for $0 \le m \le n$, the amplitude carrying the exponent Δ being the one attached to the *lower* index. Note that the ratio $n!/\Delta!$ arises whole from the Pochhammer reduction and therefore sits *outside* the square root. We verify Eq. (76) against direct Fock summation to 2×10^{-16}.

B Cross matrix element of the number operator

Repeating the sum with each term weighted by its Fock eigenvalue $\ell = n + p$,

$$\kappa_\alpha^{mn} = \langle \alpha, m|\, a^\dagger a\, |\alpha, n\rangle = A_m A_n \alpha^\Delta \big[n\, S(x) + x\, S'(x)\big], \qquad x = \alpha^2, \tag{77}$$

where $S(x) = (n!/\Delta!)\, {}_1F_1(n+1; \Delta+1; x)$ is the sum of Appendix A. Using $\frac{d}{dx}{}_1F_1(a; b; x) = \frac{a}{b}{}_1F_1(a+1; b+1; x)$ together with a Kummer contiguous relation, this reduces to

$$\kappa_\alpha^{mn} = g_{mn}(\alpha)\, Q_{mn}(x), \qquad Q_{mn}(x) = (n+1)\frac{{}_1F_1(n+2; \Delta+1; x)}{{}_1F_1(n+1; \Delta+1; x)} - 1. \tag{78}$$

Being a dimensionless ratio, Q_{mn} is insensitive to the prefactor conventions of Appendix A. Verified against Fock summation to 6×10^{-15}.

C Numerical conventions

All results use real amplitudes. Balanced plots take $\mu = \nu = 1/\sqrt{2}$; two-dimensional weight maps impose $\nu = \sqrt{1-\mu^2}$ with $0 \le \mu \le 1$. Uhlmann plots show the principal value $\operatorname{Arg} Z_U \in (-\pi, \pi]$. The shaded region above α_g in Figs. 6 and 8 is analytic, not a numerical cutoff: it is where $p = 1/2$ has no solution. Root-finding uses bisection on the monotonic $|g_{mn}(\alpha)|^2$; Laguerre polynomials and ${}_1F_1(a; b; z)$ are evaluated with `scipy.special`. The truncated Fock space uses 90 levels per mode, ample for $\alpha \lesssim 3$. The complete code accompanies this manuscript as `pacs_global_cycle.ipynb`.

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